

Started on	Sunday, 12 February 2023, 8:51 PM
State	Finished
Completed on	Sunday, 12 February 2023, 8:51 PM
Time taken	11 secs
Marks	0/38
Grade	0 out of 100

Question 1

Not answered

Marked out of 1

A meta-analysis shows favourable results for a drug A compared to drug B. Number of trials favouring drug B that needs to exist in order to negate the results shown by the metaanalysis is calculated by

Select one:

- ☐ Galbraith plot
- ☐ Receiver operator curve
- ☐ Q statistics
- ☐ Funnel plot
- ☐ Failsafe N

Your answer is incorrect.

Researchers often choose not to publish results that are not significant and store them in their file drawers! This leads to bias in the results of a meta-analysis. Rosenthal proposed a measure to estimate the number of non-significant articles remaining in file drawers by testing how many additional results would it take to reduce the overall significant results of a meta-analysis to non-significance. This is called failsafe N.

The correct answer is: Failsafe N

Question 2

Not answered

Marked out of 1

A meta-analysis of antipsychotic related diabetes suggests that first-generation antipsychotics reduce the relative risk of diabetes by around 25% in comparison to second generation antipsychotics. You have a 32 year old man requiring antipsychotic prescription; with BMI <25 and no family history of diabetes, the risk of developing diabetes in next one year may be as low as 0.5% in this patient. How many such patients must be given first generation antipsychotic instead of second generation agent to prevent one case of diabetes

Select one:

- ☐ 2
- ☐ 800
- ☐ 100
- ☐ 25
- ☐ 125

Your answer is incorrect.

This individual's risk of diabetes is 0.5%. In comparison to SGA, FGA would reduce this risk by 25% to 0.375%. The ARR is 0.125%. The inverse of this ARR (that is, 100 divided by the ARR expressed as a percentage) is equal to the number of such patients we would have to treat to prevent 1 event (in this case, to prevent 1 new onset diabetes), the NNT. In this case, we would have to treat approximately 800 such patients ($100/0.125 = 800$). You can also do this sum in a simpler way. $NNT = 1 / (RRR \times PEER)$. Here, PEER, is 0.5%, RRR, reduction in relative risk is 25%. $1 / (25\% \times 0.5\%) = 1 / 0.125\% = 100 / 0.125 = 800$. Smith M, Hopkins D, Peveler RC, Holt RI, Woodward M, Ismail K. First- v. second-generation antipsychotics and risk for diabetes in schizophrenia: systematic review and meta-analysis. Br J Psychiatry 2008; 192: 406 -11

The correct answer is: 800

Question 3

Not answered

Marked out of 1

A meta-analysis of antipsychotic related diabetes suggests that first generation antipsychotics reduce the relative risk of diabetes by around 25% in comparison to second generation antipsychotics. You have a 40-year-old Afro-Caribbean man requiring antipsychotic prescription; with BMI of 29 and positive family history of diabetes, the risk of developing diabetes in next one year is estimated to be 20% in this patient. How many such patients must be given first generation antipsychotic instead of second generation agent to prevent one case of diabetes

Select one:

- ☐ 20
- ☐ 125
- ☐ 100
- ☐ 2
- ☐ 25

Your answer is incorrect.

This individual's risk of diabetes is 20%. In comparison to SGA, FGA would reduce this risk by 25% to 15%. The ARR is 5%. The inverse of this ARR (that is, 100 divided by the ARR expressed as a percentage) is equal to the number of such patients we would have to treat to prevent 1 event (in this case, to prevent 1 new onset diabetes), the NNT. In this case, we would have to treat approximately 20 such patients ($100/5 = 20$). Smith M, Hopkins D, Peveler RC, Holt RI, Woodward M, Ismail K. First- v. second-generation antipsychotics and risk for diabetes in schizophrenia: systematic review and meta-analysis. Br J Psychiatry 2008; 192: 406 -11

The correct answer is: 20

Question 4

Not answered

Marked out of 1

The outcome measure that varies most with control event rate is

Select one:

- ☐ Experimental event rate
- ☐ Absolute risk reduction
- ☐ Relative risk reduction
- ☐ Relative risk
- ☐ Odds ratio

Your answer is incorrect.

A study of 115 meta-analyses showed that the risk difference (ARR) varied most over different populations, whereas the RR and OR were about equally stable (Schmid et al. 1995). As RR is, stable one would expect RRR also to be stable. EER need not change with CER.

The correct answer is: Absolute risk reduction



Question 5

Not answered

Marked out of 1

In a systematic review of 34 eligible studies comparing the efficacy of quetiapine in bipolar depression against placebo, a meta-analysis is reported to be inaccurate due to heterogeneity. All of the following methods can be used to identify statistical heterogeneity except

Select one:

- ☐ Plotting a measure of precision against standardised estimates
- ☐ Cochrane Chi square test
- ☐ Using a forest plot
- ☐ Using a second independent reviewer to select eligible studies
- ☐ Plotting the event rate in the treatment group on the vertical axis and in the control group on the horizontal axis

Your answer is incorrect.

Meta-analysts typically use two statistical approaches to evaluate the extent of variability in results between studies (i.e., heterogeneity) - Cochran's Q test and the I² statistic. The basic intent of a meta-analysis is to include enough studies to narrow the confidence interval around the resulting pooled estimate. There are 2 criteria for not combining the results of studies in a meta-analysis: highly disparate point estimates and confidence intervals with little overlap - both these will result in a pooled estimate with wider confidence interval that beats the primary aim. Examining a forest plot, using Galbraith plot (choice B) and L'Abbe plot (Choice C) can also help estimating qualitatively the extent of heterogeneity. CMAJ MAR. 1, 2005; 172 (5)

The correct answer is: Using a second independent reviewer to select eligible studies

Question 6

Not answered

Marked out of 1

Which of the following components of a forest plot depict the weight of a study in a meta-analysis?

Select one:

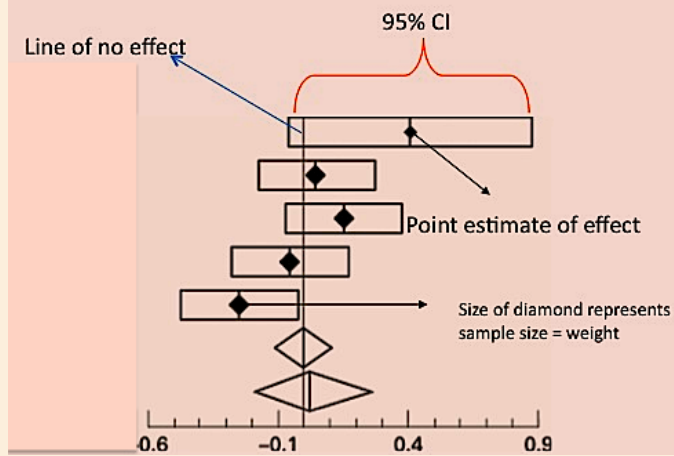
- ☐ Rank order of the study among the list in a forest plot
- ☐ The width of horizontal line corresponding to a study
- ☐ The position of vertical line of no effect relative to the study
- ☐ Size of black square representing the study
- ☐ Weight cannot be determined by visual inspection

Your answer is incorrect.

See the attached image of a forest plot



Forest plot



The correct answer is: Size of black square representing the study

Question 7

Not answered

Marked out of 1

A meta-analysis summarises ten studies and reports a significant Cochran Q test result. Which of the following is the correct interpretation

Select one:

- ☐ Significant homogeneity exists among the reported studies
- ☐ Methodological quality of individual studies are significantly inferior to summary results
- ☐ The summary estimate is statistically significant
- ☐ Significant heterogeneity exists among the reported studies
- ☐ Significant publication bias exists

Your answer is incorrect.

Cochran's Q test is the traditional test for heterogeneity. It tests the null hypothesis that all of the apparent variability is due to chance (i.e. no statistical heterogeneity is present). The test is in a way a modified test of two probability distribution; if the test generates a low p-value it means the null hypothesis is rejected i.e. statistical heterogeneity exists. If the p-value is low (say, less than 0.1) investigators should look for possible explanations including differences in patients profile, interventions, measurement of outcomes and study design. If the p-value gets very low (less than 0.01) one must avoid or reconsider the use of pooled single best estimates of treatment effects. But this test is sensitive to the sample size of individual studies - when studies have included only a few patients it may be difficult to reject the null hypothesis even if it is false. On the other hand, when sample sizes are very large, even small differences in effect size that may not be clinically meaningful, may nevertheless generate low p-values.

The correct answer is: Significant heterogeneity exists among the reported studies

Question 8

Not answered

Marked out of 1

A meta-analysis was carried out for studies analysing the use of hormone replacement and incidence of depression in postmenopausal women. Using a random effects model it was noted that the relative risk of depression in those receiving hormone replacement therapy was $RR = 0.55$ (HRT was beneficial). There was substantial heterogeneity between studies. Which of the following is true from the above study

Select one:

- ☐ Individual estimates were close to each other.
- ☐ Degree of heterogeneity cannot be quantified.
- ☐ A random effects model was used because there was heterogeneity between studies.
- ☐ All of the above included trials showed beneficial effect for HRT individually.
- ☐ The confidence interval of summary statistics would have been narrower than the confidence interval generated using fixed effects model.

Your answer is incorrect.

Degree of heterogeneity can be quantified using statistical methods - it is difficult to quantify the degree of homogeneity on the other hand. Random effects model is used due to the presence of heterogeneity - as a result, the pooled estimate may have wider confidence limits.

The correct answer is: A random effects model was used because there was heterogeneity between studies.

Question 9

Not answered

Marked out of 1

Which of the following is the most important assumption made when choosing fixed effect analysis in a systematic review

Select one:

- ☐ Difference in outcome of individual studies is due to geographical differences
- ☐ No difference is noted between the outcome of individual studies
- ☐ Difference in outcome of individual studies is due to different scales used
- ☐ Difference in outcome of individual studies is due to chance
- ☐ Difference in outcome of individual studies is due to variations between studies

Your answer is incorrect.

The usual method of meta-analysis assumes that all the studies are estimating the same effect and only random variation between research subjects within a study leads the observed study effects to vary. We call this the fixed effects model. Here we assume that the effect to be estimated is the same in all studies and use only the sampling variation within the studies. In other words, we assume that any difference between studies is due to chance. When one cannot explain why different studies exploring the same phenomenon throw up different results and estimates, we can use the random effects model to arrive at pooled estimate. This means we are accepting the existence of the heterogeneity and allow for the degree of uncertainty that results while calculating summary effects. When using a random effects model, we regard each study as estimating a different effect in a population. We estimate both the variability between subjects within the studies (consider this to be equivalent to sampling variation) and the variation between studies (population variability). The other stated differences pertain to methodological variability. Tyrer 48

The correct answer is: Difference in outcome of individual studies is due to chance

Question 10

Not answered

Marked out of 1

In a meta-analysis compiling outcome measure of continuous data measured using different rating scales, which of the following measure can be used to synthesize results

Select one:

- ☐ Weighted relative risk
- ☐ Standardised mean difference
- ☐ Weighted odds ratio
- ☐ Weighted mean difference
- ☐ Weighted kappa

Your answer is incorrect.

In a meta-analysis, the results of individual studies are combined to give one overall estimate of treatment effect if the patients, interventions and outcomes in the various studies are sufficiently similar. However, if the studies addressed the same conceptual entity (e.g. depression) but applied different instruments or scales, the results cannot be combined directly. In such circumstances, the result of each individual study can be standardized by calculating a standardized mean difference (SMD) which is the difference of the means of both treatment arms divided by the pooled standard deviation. This allows the treatment effect in each study to be expressed in standard units (rather than the original units), and the results of all studies can be combined. Gotzsche, P. C., Hrobjartsson, A., Maric, K., et al. (2007) Data Extraction Errors in Meta-analyses That Use Standardized Mean Differences. JAMA, 298, 430-437

The correct answer is: Standardised mean difference

Question 11

Not answered

Marked out of 1

In a meta-analysis compiling outcome measure of continuous data measured using same rating scale, which of the following measure can be used to synthesize results

Select one:

- ☐ Weighted relative risk
- ☐ Weighted mean difference
- ☐ Standardised mean difference
- ☐ Weighted odds ratio
- ☐ Weighted kappa

Your answer is incorrect.

Results from trials that have measured the same outcome on the same scale can readily be combined in a meta-analysis by calculating the weighted mean difference. Sometimes, trials have used outcomes of a similar nature but that were measured on different scales, e.g., depression on a clinician-rated scale such as the Hamilton Rating Scale for Depression or a self-rating scale such as the Beck Depression Inventory. In such cases, it is necessary to standardize the measurements on a uniform scale before they can be pooled in a meta-analysis. This is done by calculating the standardized mean difference (SMD) for each trial, which is the difference in means between the two groups, divided by the pooled standard deviation of the measurements. By this transformation, the outcome becomes dimensionless, and the scales become uniform. Gotzsche, P. C., Hrobjartsson, A., Maric, K., et al. (2007) Data Extraction Errors in Meta-analyses That Use Standardized Mean Differences. JAMA, 298, 430-437

The correct answer is: Weighted mean difference

Question 12

Not answered

Marked out of 1

While conducting a systematic review, a researcher comes across few studies whose mean outcome observed is smaller than the standard deviation of the observations. Which of the following can be inferred from the above

Select one:

- ☐ There has been a statistical error
- ☐ The results are misappropriated in the individual studies
- ☐ Meta-analysis cannot be carried out
- ☐ There is a major publication bias
- ☐ The data from individual studies are skewed

Your answer is incorrect.

If data have a very skewed distribution, then the standard deviation will be grossly inflated and may be closer or even higher than mean; in this case SD may not be a good measure of variability to use. During meta-analysis, the researcher can either use transformation of the data, such as a log transform, which will render the distribution more symmetrical or can resort to dichotomizing the outcome and using binary variables. Freeman and Tyrer, Research Methods in Psychiatric. RCPsych publication 2006; Pg 47

The correct answer is: The data from individual studies are skewed

Question 13

Not answered

Marked out of 1

A Cochrane review identified that in the short-term, risperidone was more likely to produce an improvement in the Positive and Negative Syndrome Scale (PANSS) when compared with haloperidol (n=2368, 9 RCTs, RR for not improved = 0.72; CI 0.59 to 0.88. Absolute risk difference = 0.125; CI 0.100 to 0.150). How many patients must be treated with risperidone instead of haloperidol to have one improvement?

Select one:

- ☐ 4
- ☐ 25
- ☐ 12.5
- ☐ 16
- ☐ 8

Your answer is incorrect.

This question seeks NNT. $NNT = 1/ARR$. Here $ARR = 0.125$. So $1/0.125 = 8$.

The correct answer is: 8



Question 14

Not answered

Marked out of 1

A Cochrane review identified that in the short-term, risperidone was more likely to produce an improvement in the Positive and Negative Syndrome Scale (PANSS) when compared with haloperidol (n=2368, 9 RCTs, RR for not improved = 0.72; CI 0.59 to 0.88. Absolute risk difference = 0.125; CI 0.100 to 0.150). What is the 95% confidence interval for NNT from the above data?

Select one:

- ☐ 100 to 150
- ☐ 0 to 8
- ☐ 1 to 15
- ☐ 1 to infinity
- ☐ 7 to 10

Your answer is incorrect.

To calculate 95% CI of NNT use the 95% CI of ARR. Higher limit NNT = $1/\text{lower limit of ARR} = 1/0.1 = 10$. Lower limit of NNT = $1/\text{higher limit of ARR} = 1/0.150 = 7$ 95% CI of NNT = 7 to 10.

The correct answer is: 7 to 10

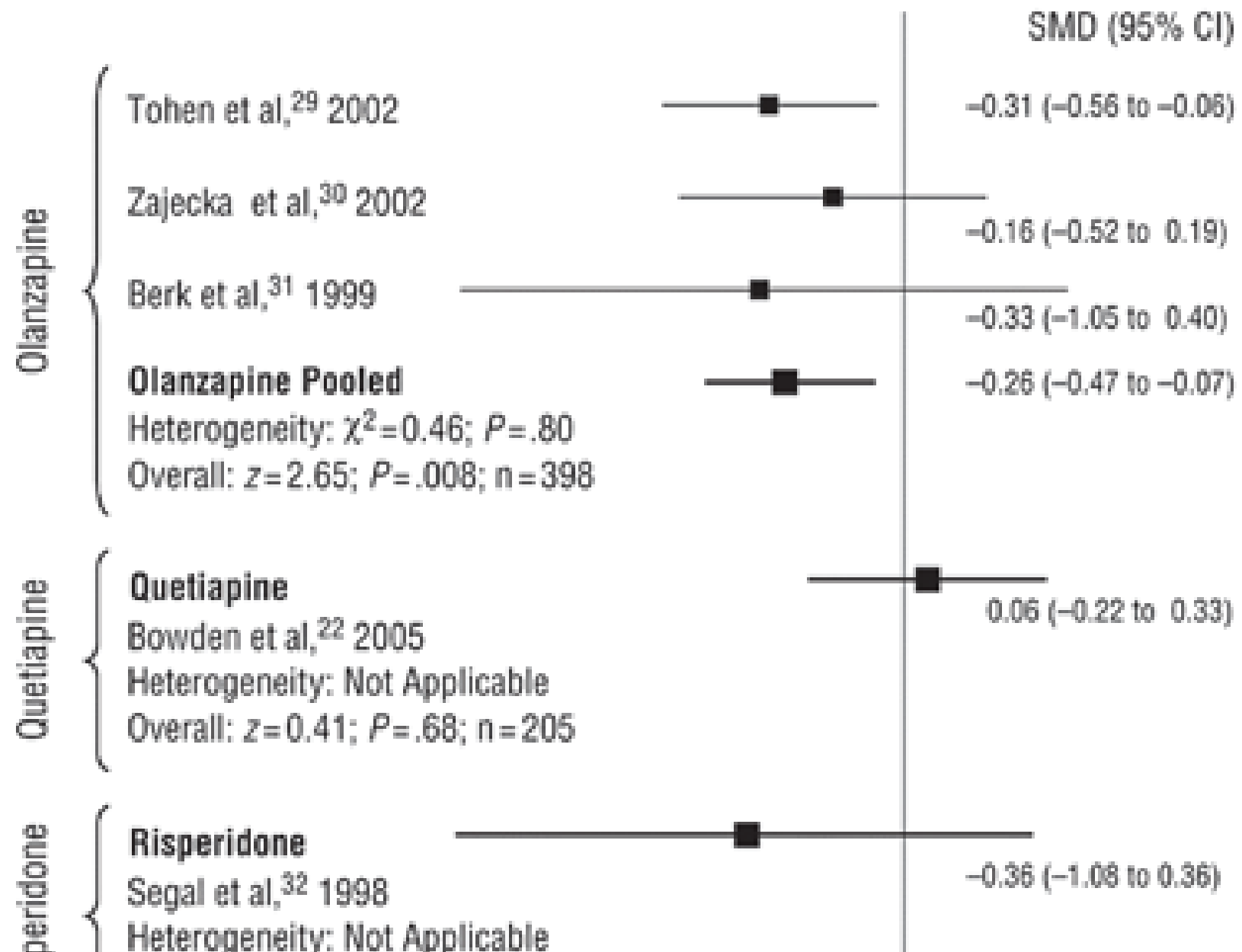
Question **15**

Not answered

Marked out of 1

Please see the below graph. Which of the following best describes the graph below?





Select one:

- ☐ Galbraith plot
- ☐ Funnel plot
- ☐ scattergram
- ☐ L'Abbe Plot
- ☐ Blobbogram

Overall: $z = 0.97$; $P = .33$; $n = 30$

All SGAs Pooled

Heterogeneity: $\chi^2 = 4.30$; $P = .37$

Overall: $z = 1.96$; $P = .05$; $n = 633$

-0.17 (-0.33 to 0.00)

Your answer is incorrect.

This graph is a forest plot - also known as a blobbogram.

The correct answer is: Blobbogram

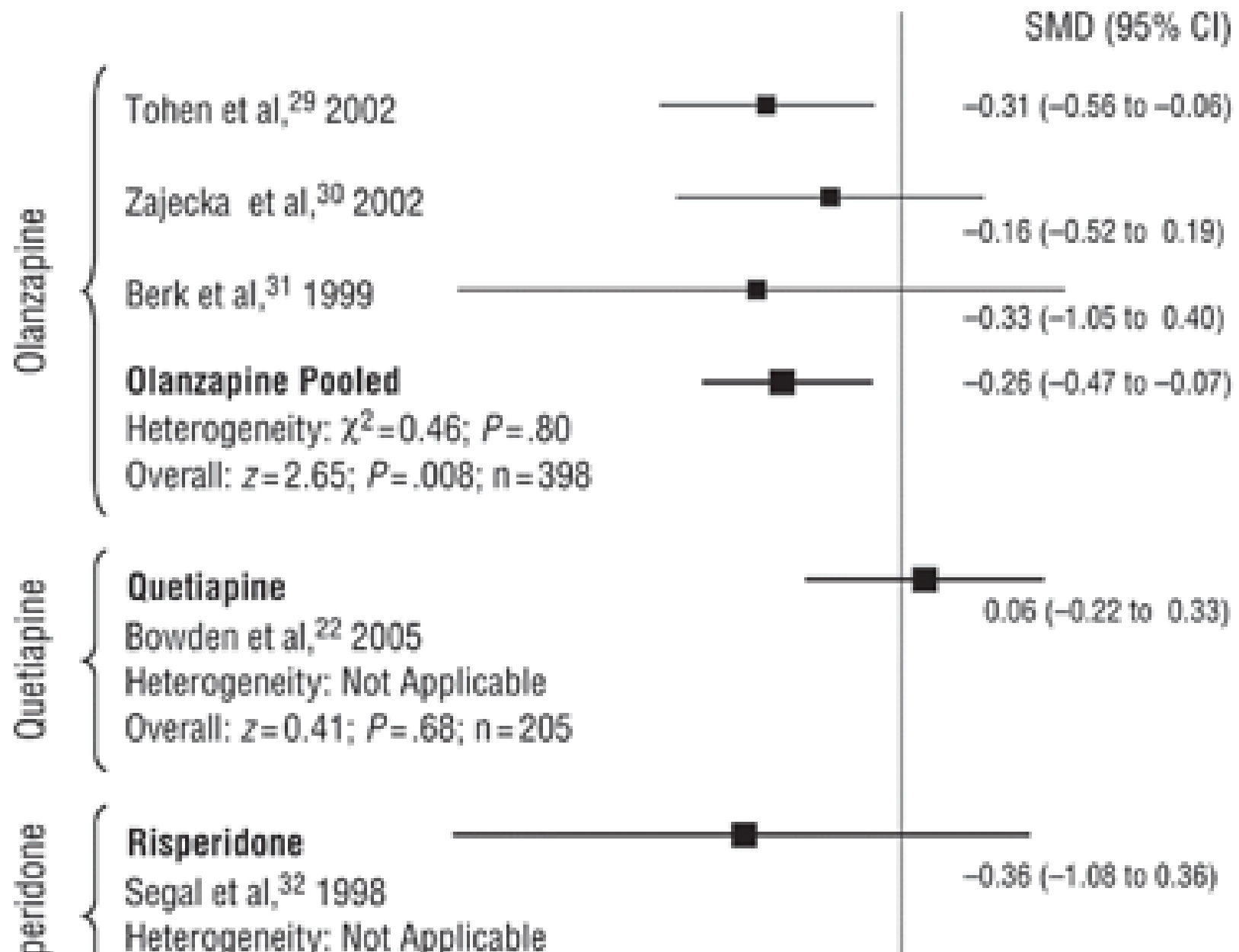
Question **16**

Not answered

Marked out of 1

Please see the below graph. What type of study has been conducted?





Select one:

- ☐ Pragmatic RCT
- ☐ A mega trial
- ☐ Secondary research
- ☐ Ecological study
- ☐ Qualitative research

Overall: $z = 0.97$; $P = .33$; $n = 30$

All SGAs Pooled

Heterogeneity: $\chi^2 = 4.30$; $P = .37$

Overall: $z = 1.96$; $P = .05$; $n = 633$

-0.17 (-0.33 to 0.00)

Your answer is incorrect.

This graph is a forest plot - also known as a blobbogram. These graphs are produced in meta-analytic studies (secondary research)

The correct answer is: Secondary research

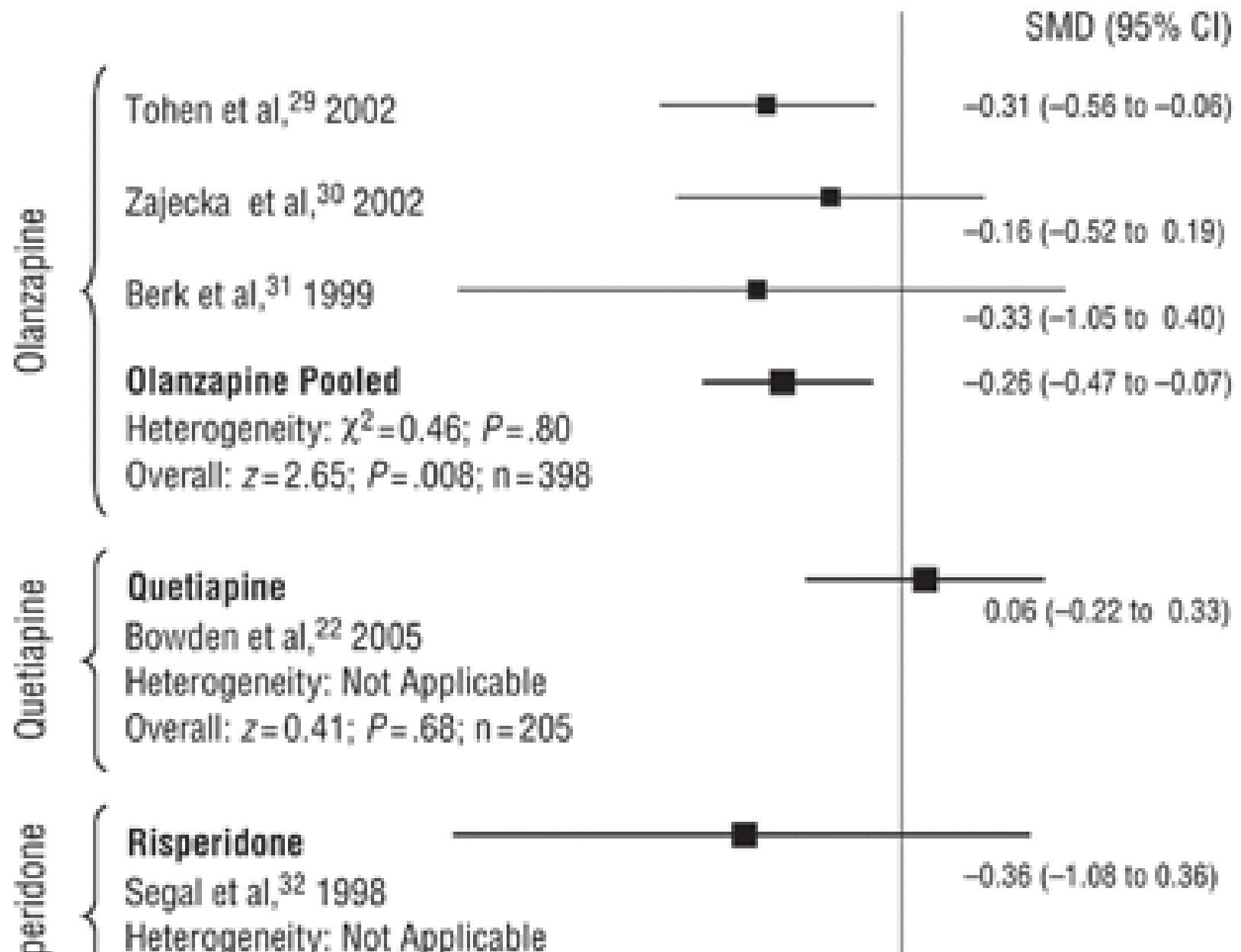
Question **17**

Not answered

Marked out of 1

Please see the below graph. Which of the following studies provides a result favouring mood stabilisers when compared to second generation antipsychotics in the intervention carried out as per the forest plot?





Select one:

- ☐ Berke et al
- ☐ Tohen et al
- ☐ Bowden et al
- ☐ Segal et al
- ☐ Zajecka et al

Overall: $z = 0.97$; $P = .33$; $n = 30$ **All SGAs Pooled**Heterogeneity: $\chi^2 = 4.30$; $P = .37$ Overall: $z = 1.96$; $P = .05$; $n = 633$ 

Your answer is incorrect.

The results favouring atypicals are on the left side of the graph, and those favouring mood stabilizers are on the right side. Only Bowden et al. study favours the mood stabilizers (even this study has confidence intervals overriding the zero point).

The correct answer is: Bowden et al



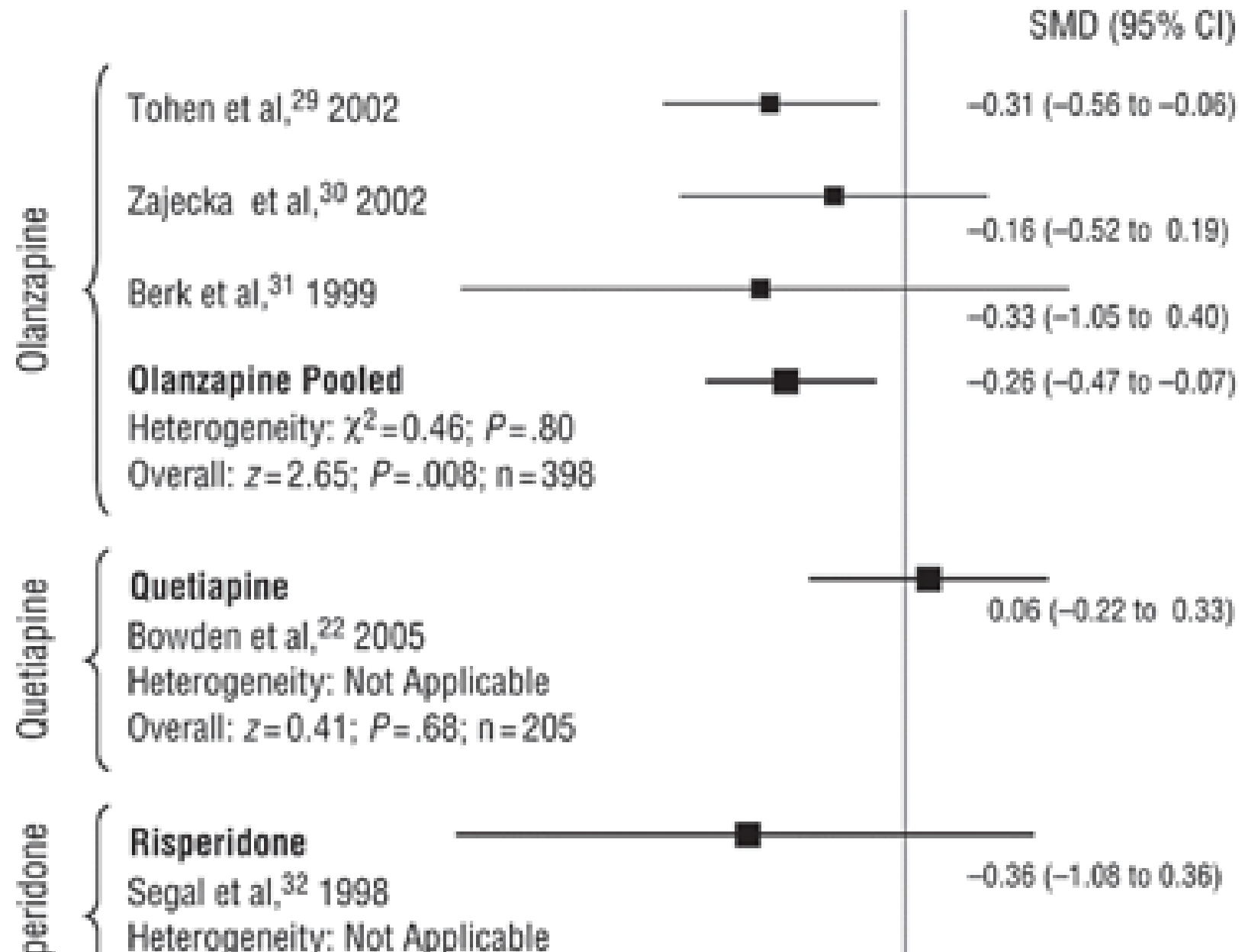
Question **18**

Not answered

Marked out of 1

Please see the below graph. If p value set for the presence of heterogeneity is $p < 0.1$, which of the following regarding heterogeneity in the above forest plot is true?





Select one:

- ☐ Heterogeneity cannot be calculated for single trials
- ☐ Statistical estimates of heterogeneity in a meta-analysis are robust tests
- ☐ Heterogeneity is not significant for Bowden trial as it favoured mood stabilisers
- ☐ Significant trial heterogeneity noted when all SGA trials are pooled
- ☐ Significant trial heterogeneity noted when olanzapine trials are pooled

Overall: $z = 0.97$; $P = .33$; $n = 30$

All SGAs Pooled

Heterogeneity: $\chi^2 = 4.30$; $P = .37$

Overall: $z = 1.96$; $P = .05$; $n = 633$

-0.17 (-0.33 to 0.00)

Your answer is incorrect.

Heterogeneity refers to the degree to which individual studies differ from each other. So by definition it cannot be calculated for a stand alone trial as Bowden et al. and Segal et al. As p is >0.1 in both olanzapine and SGAs pooled data, options A and B are incorrect.

The correct answer is: Heterogeneity cannot be calculated for single trials

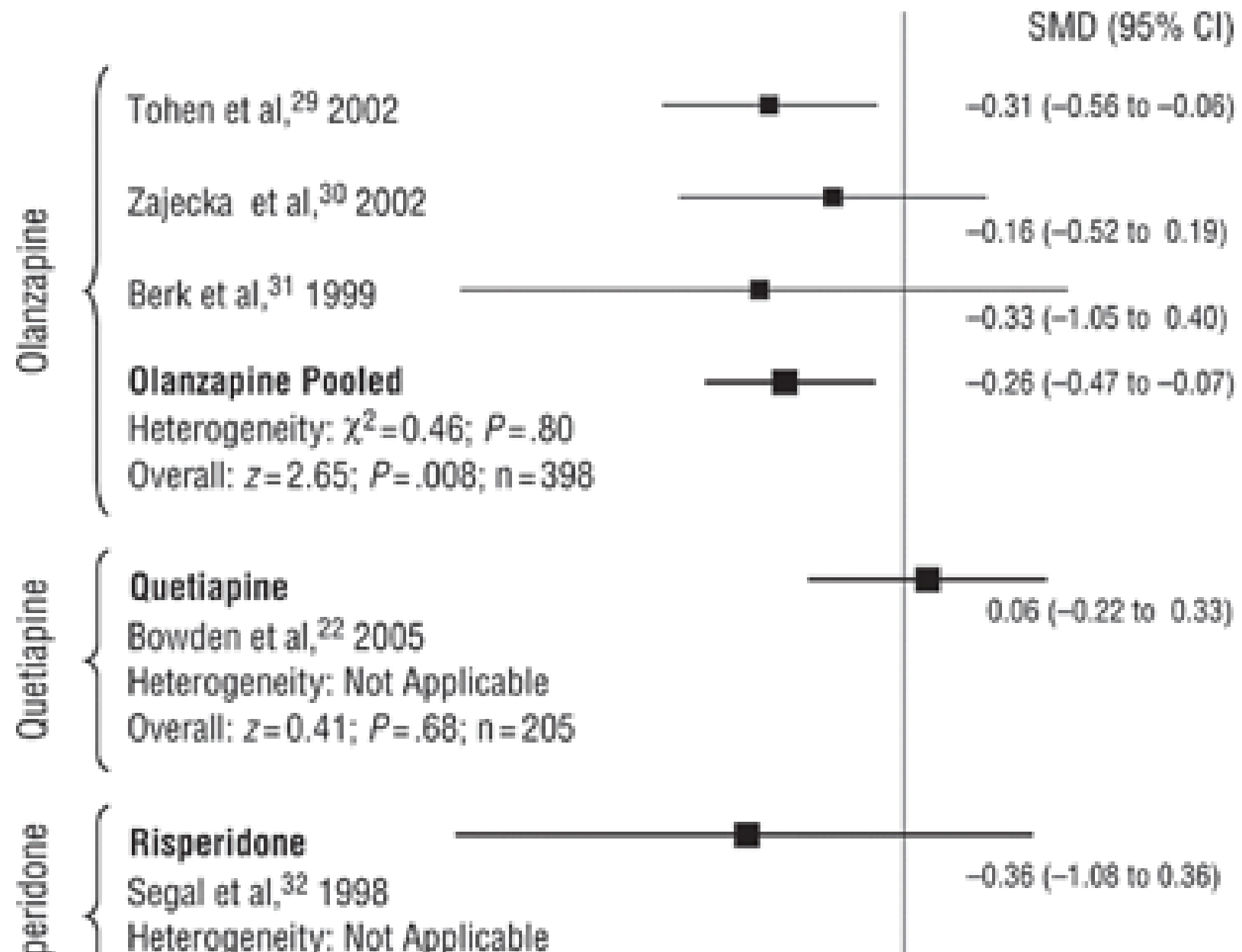
Question **19**

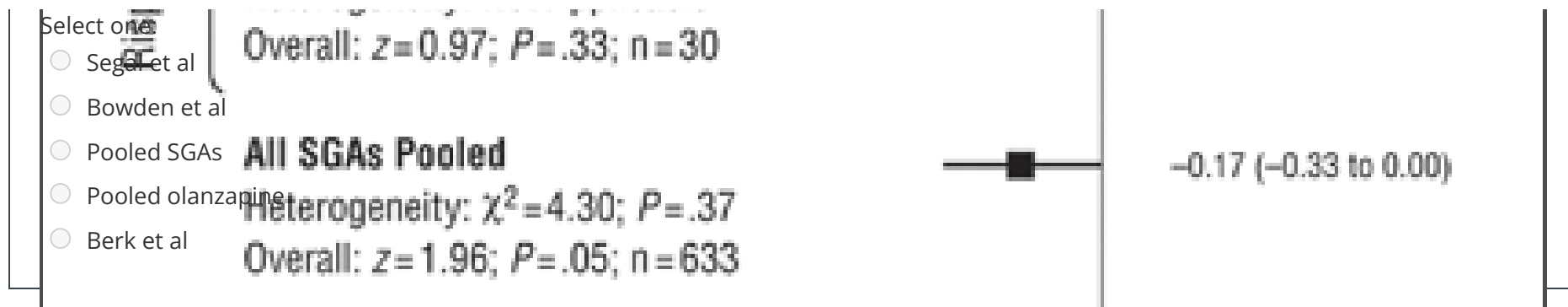
Not answered

Marked out of 1

Please see the below graph. Most precise results are noted for which one of the following?







Your answer is incorrect.

On visual inspection, the smaller horizontal lines represent all SGAs pooled, and Olanzapine pooled. On a closer look one may find that the confidence limits are wider for the latter; around 40 points compared to all SGAs pooled where it is 33 points. Hence, Pooled SGAs is the right answer.

The correct answer is: Pooled SGAs



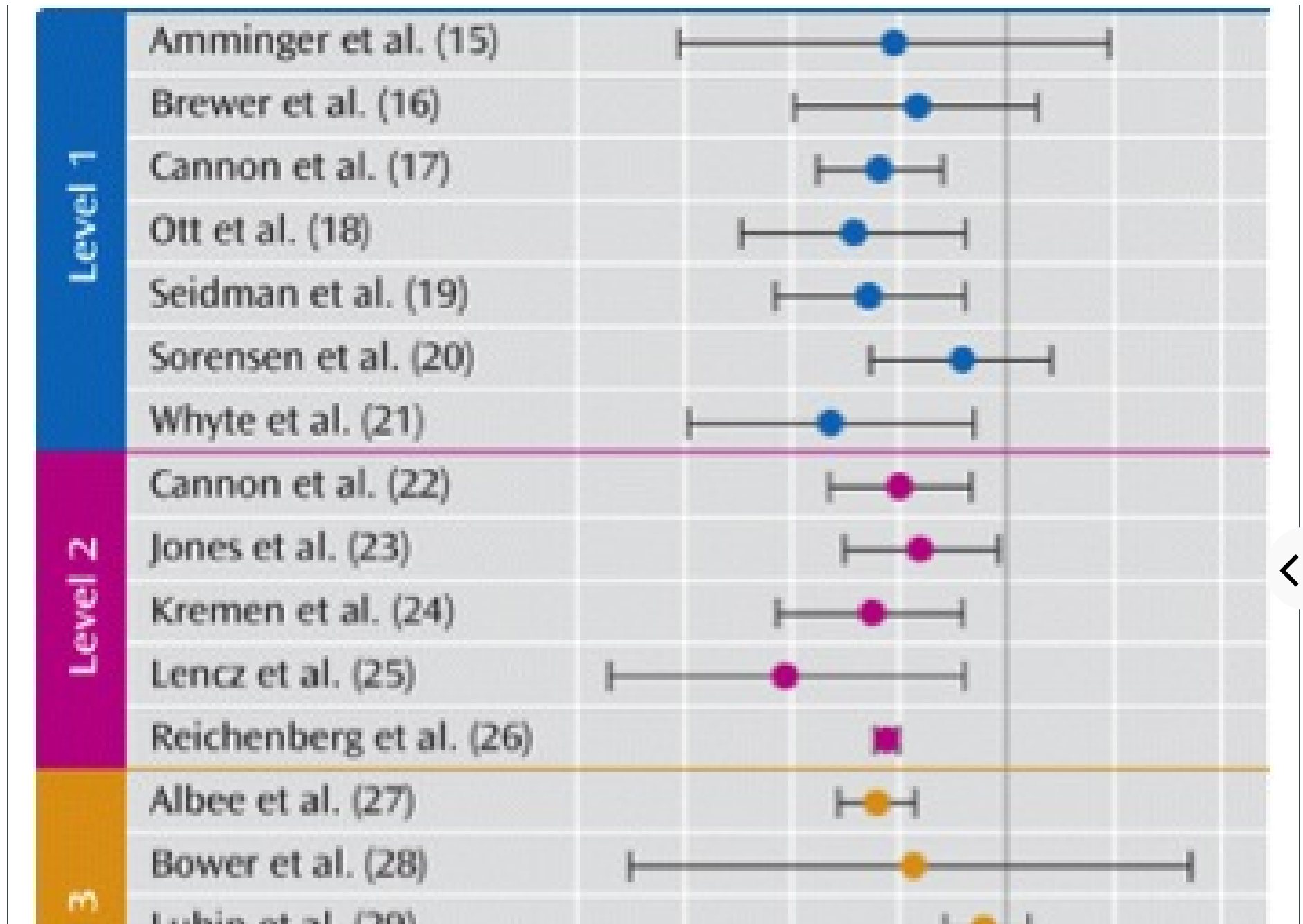
Question **20**

Not answered

Marked out of 1

Woodberry et al. (2008) investigated the available literature on premorbid IQ in schizophrenia compared to healthy controls. The mean weighted final outcome measure suggested a medium-sized deficit in global cognition prior to the onset of schizophrenia. In converting available IQ scores to a standardized score with a mean of 100 and standard deviation of 15, the mean premorbid IQ estimate for schizophrenia samples was 94.7 (0.35 standard deviations below the mean and at the lower end of the average range). In the figure, level 1 includes all studies that used recent diagnostic criteria (RDC, ICD-10, or DSM-III or later) while level 2 includes all studies that used recent diagnostic criteria (RDC, ICD-10, or DSM-III or later) and "short" IQ estimates; Level 3 includes all studies that used older diagnostic systems (pre-DSM-III or RDC) and "short" IQ estimates. (Woodberry KA et al. Premorbid IQ in Schizophrenia: A Meta-Analytic Review Am J Psychiatry 2008; 165:579-587). Which of the following conclusion is correct on visual inspection of the below figure?





Select one:

- ☐ The studies are homogenous in their reported effect sizes
- ☐ Homogeneity is noted when effect sizes within level 3 are considered
- ☐ Significant heterogeneity is noted in the reported effect sizes of individual studies
- ☐ No significant heterogeneity between methodological levels is found
- ☐ Significant heterogeneity is noted in the reported effect sizes of each levels

Your answer is incorrect.

We cannot say significant heterogeneity is present by a simple visual analysis: we need statistics for measuring the presence of significance. We cannot conclude homogeneity - we can only infer degree of heterogeneity

The correct answer is: No significant heterogeneity between methodological levels is found



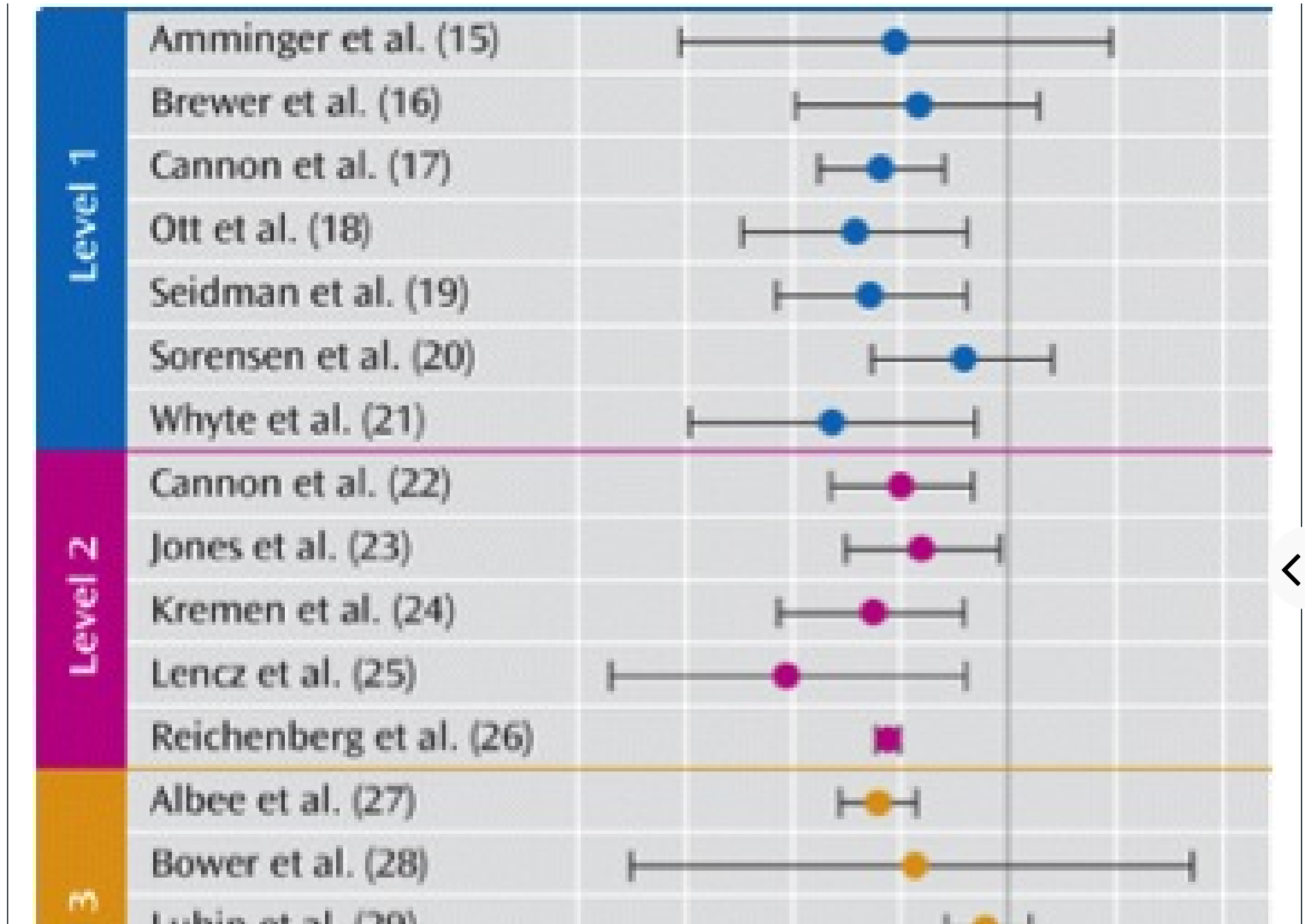
Question **21**

Not answered

Marked out of 1

Woodberry et al. (2008) investigated the available literature on premorbid IQ in schizophrenia compared to healthy controls. The mean weighted final outcome measure suggested a medium-sized deficit in global cognition prior to the onset of schizophrenia. In converting available IQ scores to a standardized score with a mean of 100 and standard deviation of 15, the mean premorbid IQ estimate for schizophrenia samples was 94.7 (0.35 standard deviations below the mean and at the lower end of the average range). In the figure, level 1 includes all studies that used recent diagnostic criteria (RDC, ICD-10, or DSM-III or later) while level 2 includes all studies that used recent diagnostic criteria (RDC, ICD-10, or DSM-III or later) and "short" IQ estimates; Level 3 includes all studies that used older diagnostic systems (pre-DSM-III or RDC) and "short" IQ estimates. (Woodberry KA et al. Premorbid IQ in Schizophrenia: A Meta-Analytic Review Am J Psychiatry 2008; 165:579-587). Which of the following statistical measure is most likely to be used to produce the below figure?





Select one:

- ☐ Cohen's kappa
- ☐ Spearman's rho
- ☐ Cohen's d
- ☐ Student's t
- ☐ Pearson's r

Lubin et al. (29)

Offord (30)

Watt and Lubensky (31)

Zammit et al. (32)

Your answer is incorrect.

Cohen's d is a measure of effect size. It is obtained by using the data from each study that enters a meta-analysis.

The correct answer is: Cohen's d

Weighted

-2.0 -1.5 -1.0 -0.5 0.0 0.5 1.0

Mean ($\pm 95\%$ CI) Effect Size

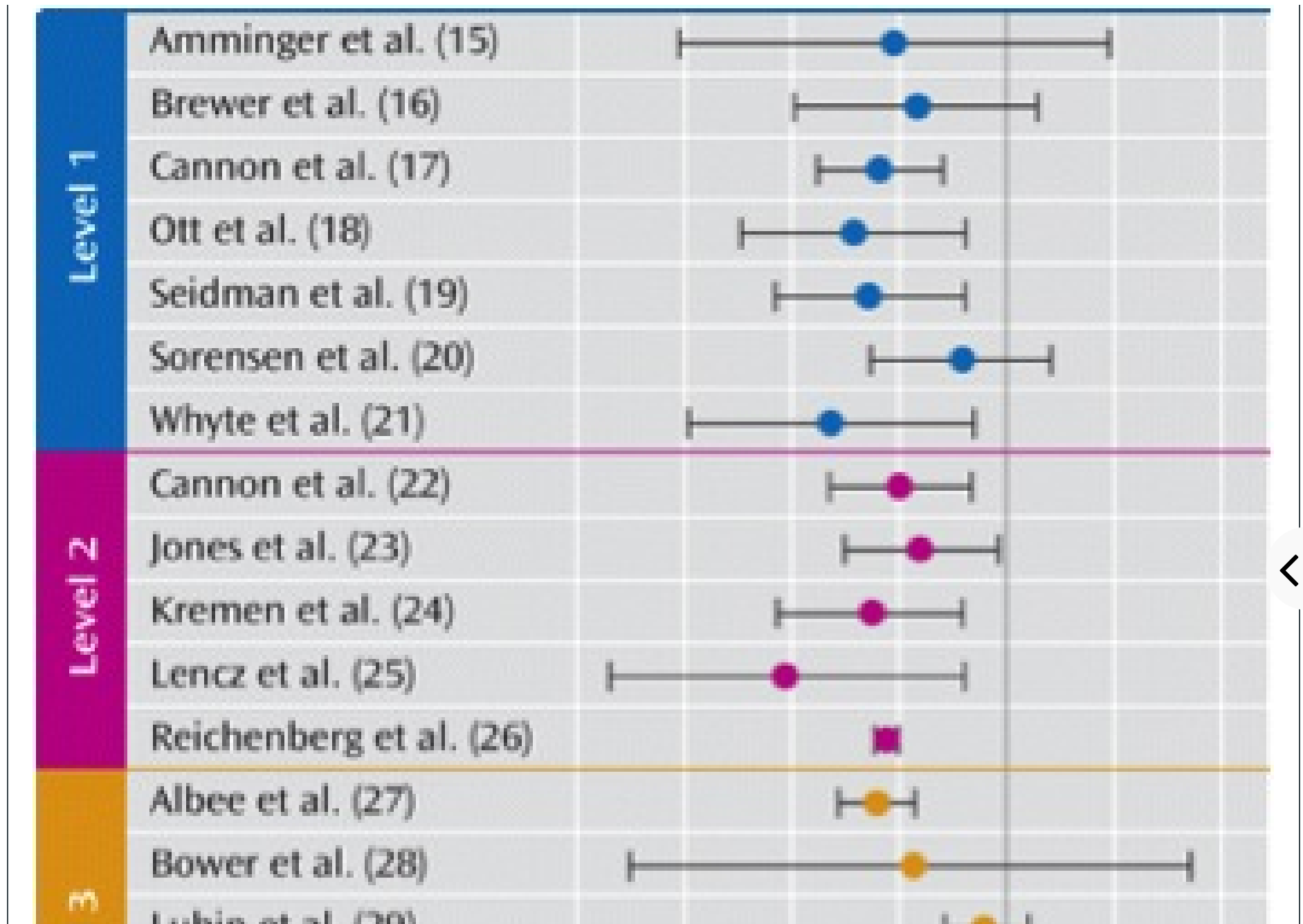
Question **22**

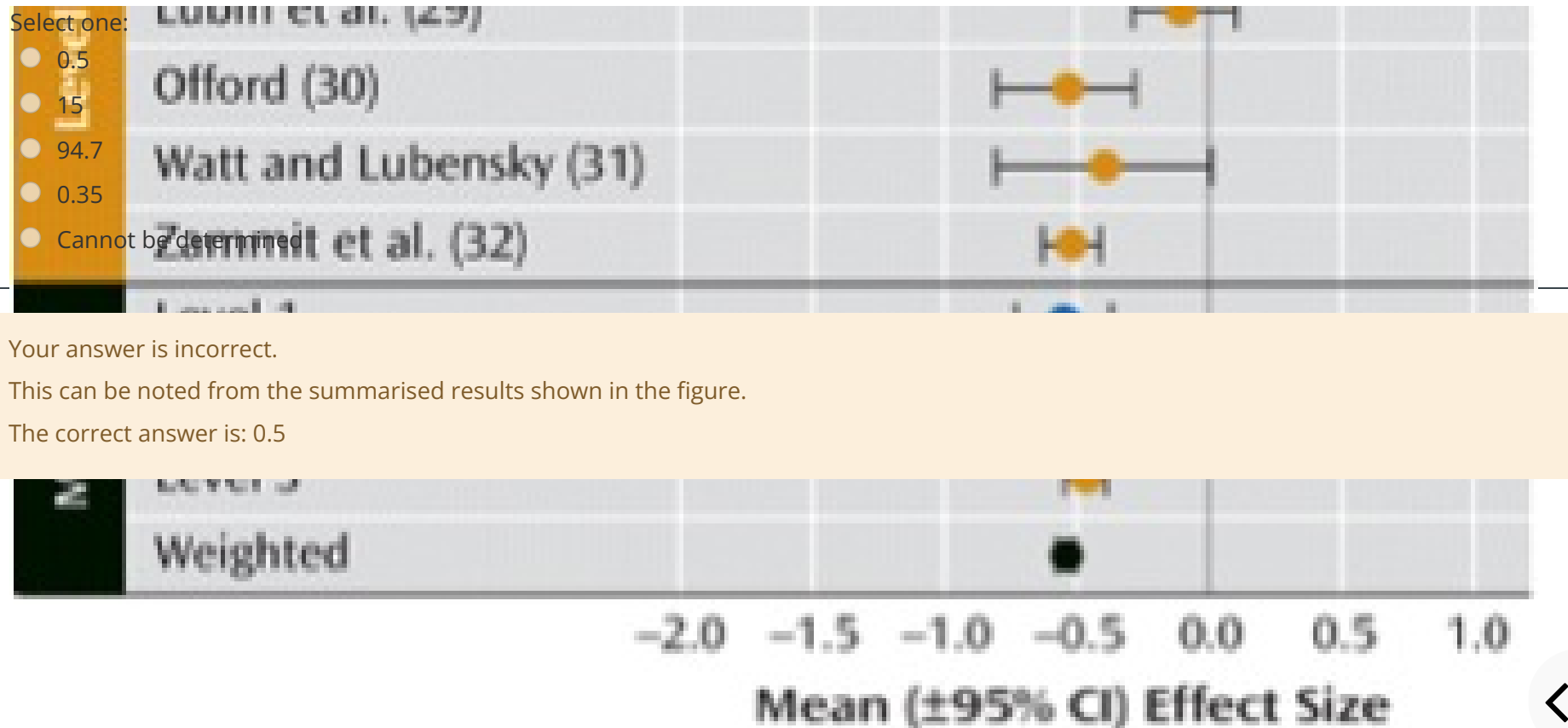
Not answered

Marked out of 1

Woodberry et al. (2008) investigated the available literature on premorbid IQ in schizophrenia compared to healthy controls. The mean weighted final outcome measure suggested a medium-sized deficit in global cognition prior to the onset of schizophrenia. In converting available IQ scores to a standardized score with a mean of 100 and standard deviation of 15, the mean premorbid IQ estimate for schizophrenia samples was 94.7 (0.35 standard deviations below the mean and at the lower end of the average range). In the figure, level 1 includes all studies that used recent diagnostic criteria (RDC, ICD-10, or DSM-III or later) while level 2 includes all studies that used recent diagnostic criteria (RDC, ICD-10, or DSM-III or later) and "short" IQ estimates; Level 3 includes all studies that used older diagnostic systems (pre-DSM-III or RDC) and "short" IQ estimates. (Woodberry KA et al. Premorbid IQ in Schizophrenia: A Meta-Analytic Review Am J Psychiatry 2008; 165:579-587). Schizophrenia samples demonstrated a medium-sized impairment in premorbid IQ. The mean effect size difference is around







Your answer is incorrect.

This can be noted from the summarised results shown in the figure.

The correct answer is: 0.5

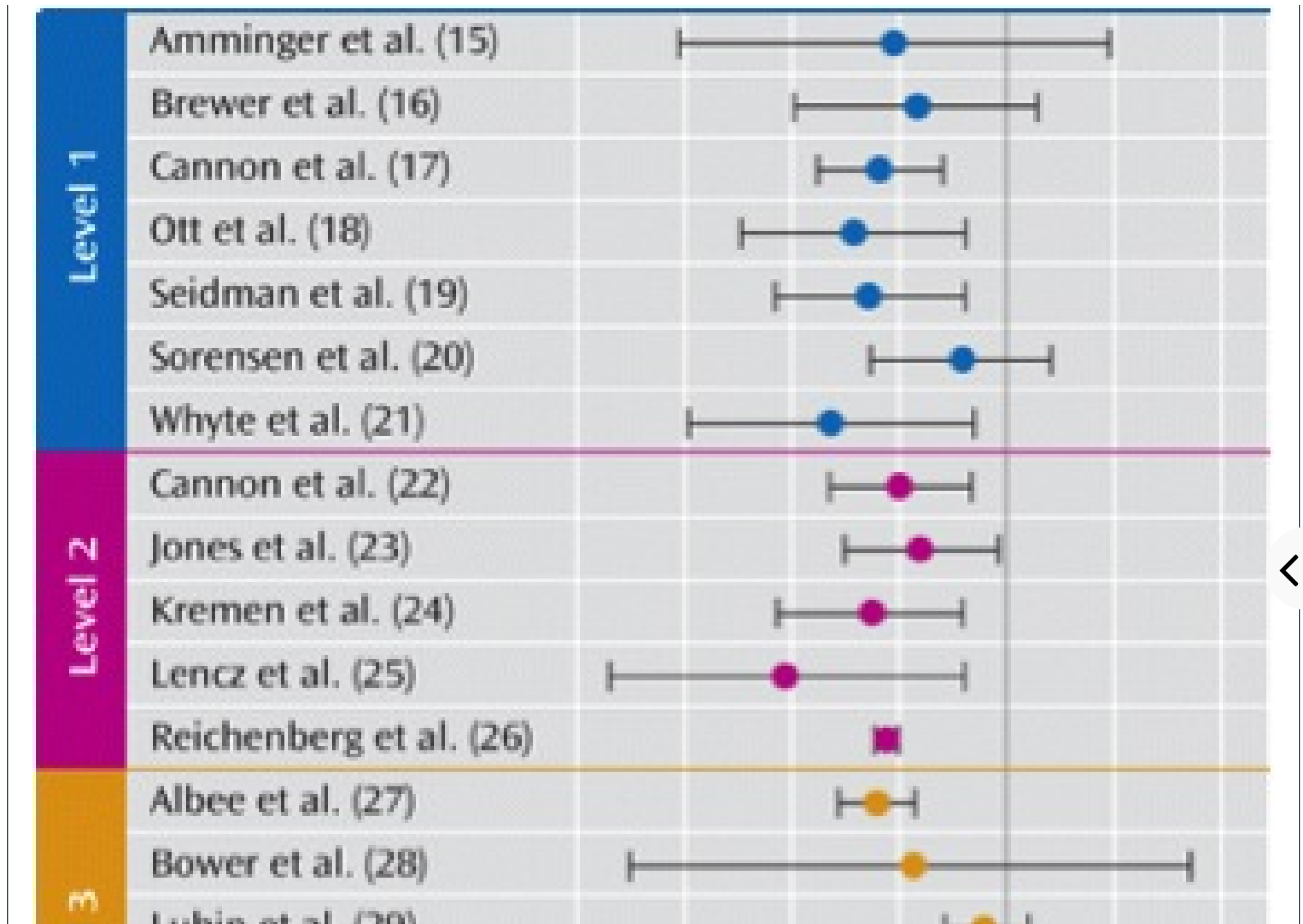
Question **23**

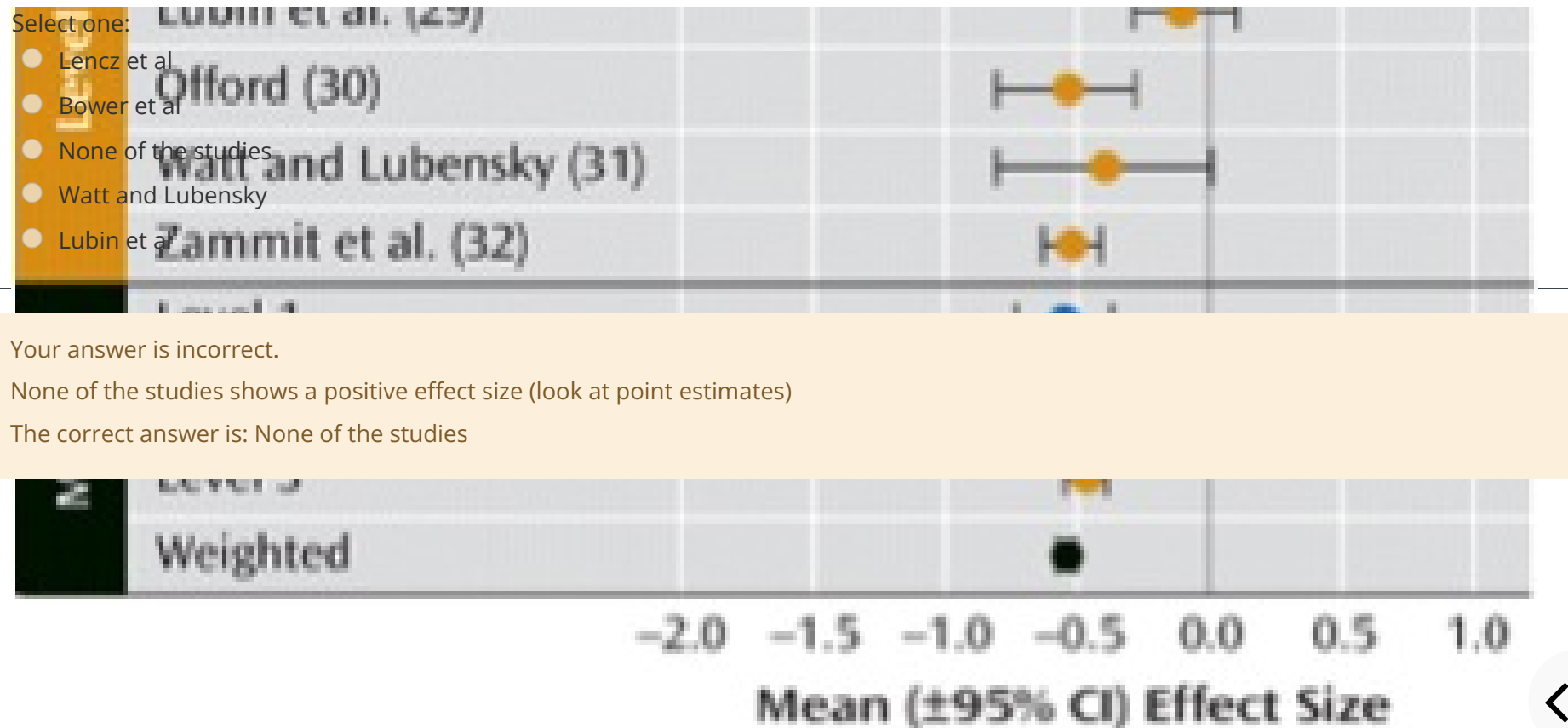
Not answered

Marked out of 1

Woodberry et al (2008) investigated the available literature on premorbid IQ in schizophrenia compared to healthy controls. The mean weighted final outcome measure suggested a medium-sized deficit in global cognition prior to the onset of schizophrenia. In converting available IQ scores to a standardized score with a mean of 100 and standard deviation of 15, the mean premorbid IQ estimate for schizophrenia samples was 94.7 (0.35 standard deviations below the mean and at the lower end of the average range). In the figure, level 1 includes all studies that used recent diagnostic criteria (RDC, ICD-10, or DSM-III or later) while level 2 includes all studies that used recent diagnostic criteria (RDC, ICD-10, or DSM-III or later) and "short" IQ estimates; Level 3 includes all studies that used older diagnostic systems (pre-DSM-III or RDC) and "short" IQ estimates. (Woodberry KA et al. Premorbid IQ in Schizophrenia: A Meta-Analytic Review Am J Psychiatry 2008; 165:579-587). The only study that reports a higher mean IQ in schizophrenia samples than healthy is







Your answer is incorrect.

None of the studies shows a positive effect size (look at point estimates)

The correct answer is: None of the studies

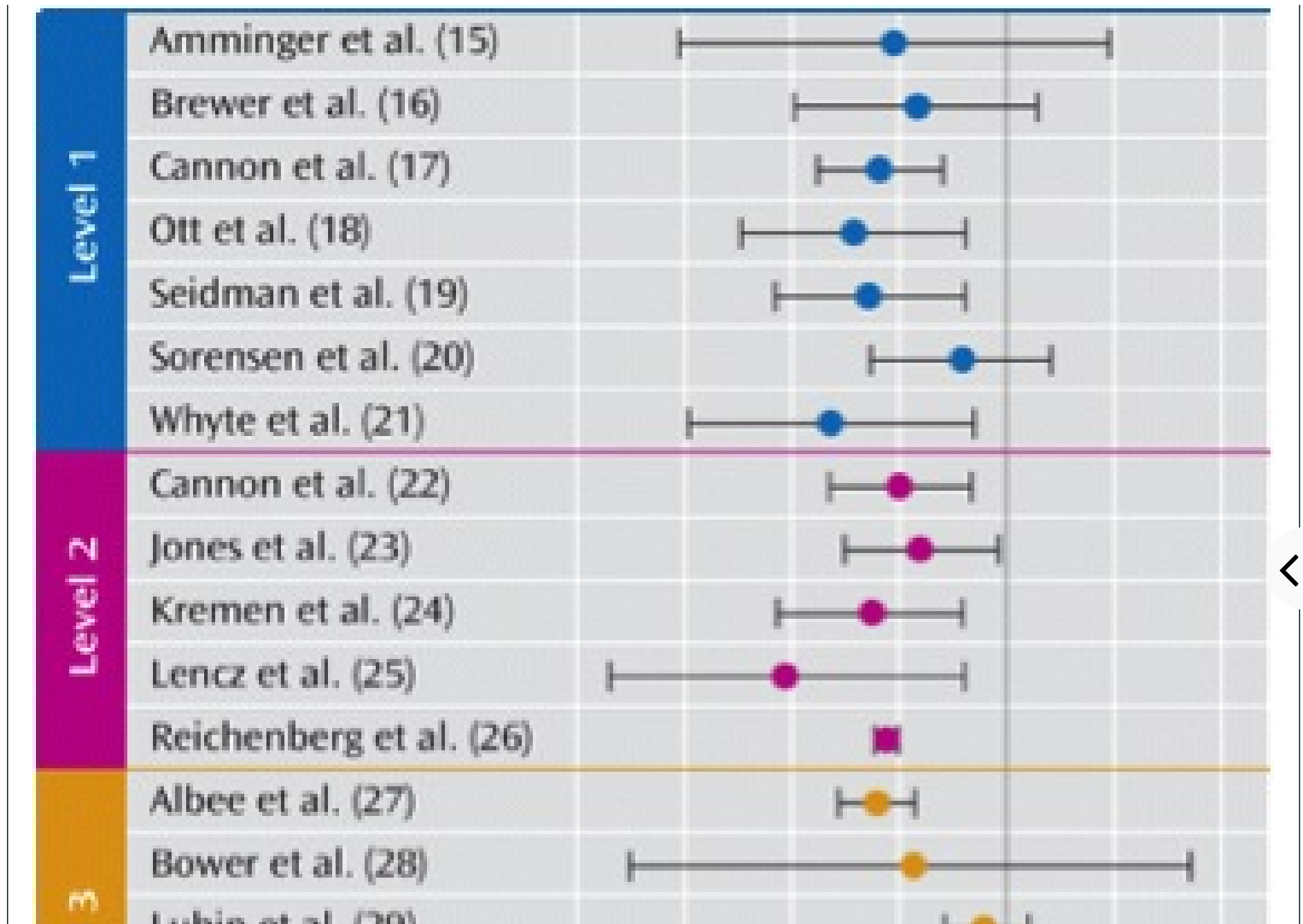
Question **24**

Not answered

Marked out of 1

Woodberry et al. (2008) investigated the available literature on premorbid IQ in schizophrenia compared to healthy controls. The mean weighted final outcome measure suggested a medium-sized deficit in global cognition prior to the onset of schizophrenia. In converting available IQ scores to a standardized score with a mean of 100 and standard deviation of 15, the mean premorbid IQ estimate for schizophrenia samples was 94.7 (0.35 standard deviations below the mean and at the lower end of the average range). In the figure, level 1 includes all studies that used recent diagnostic criteria (RDC, ICD-10, or DSM-III or later) while level 2 includes all studies that used recent diagnostic criteria (RDC, ICD-10, or DSM-III or later) and "short" IQ estimates; Level 3 includes all studies that used older diagnostic systems (pre-DSM-III or RDC) and "short" IQ estimates. (Woodberry KA et al. Premorbid IQ in Schizophrenia: A Meta-Analytic Review Am J Psychiatry 2008; 165:579-587). The individual study with highest precision of the effect size measure is







Your answer is incorrect.

Precision can be inferred from the width of horizontal line. Of all the individual studies listed in the options, Reichenberg et al. has the shortest horizontal line, so it is likely to be the study with the most precise results.

The correct answer is: Reichenberg et al



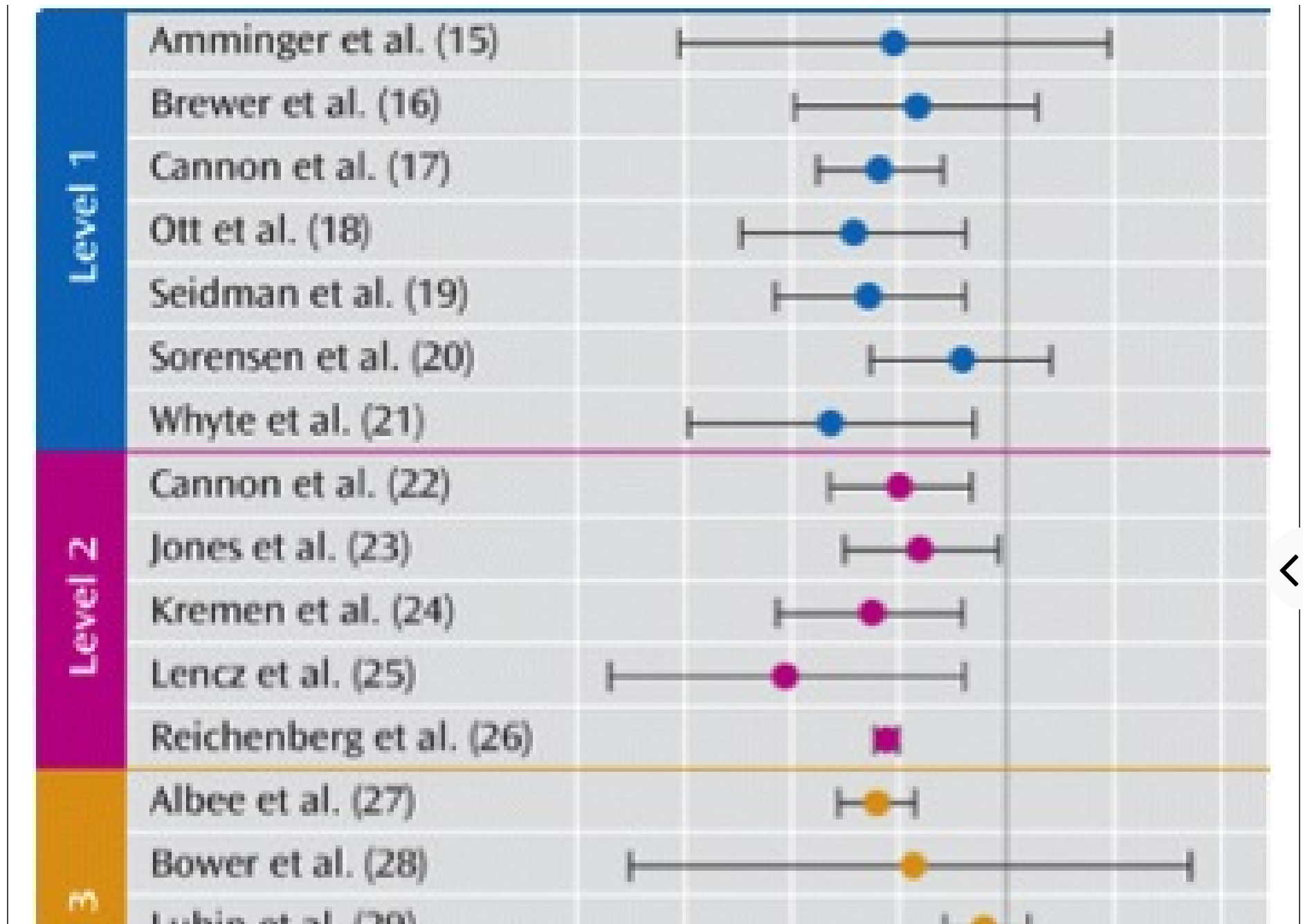
Question **25**

Not answered

Marked out of 1

Woodberry et al. (2008) investigated the available literature on premorbid IQ in schizophrenia compared to healthy controls. The mean weighted final outcome measure suggested a medium-sized deficit in global cognition prior to the onset of schizophrenia. In converting available IQ scores to a standardized score with a mean of 100 and standard deviation of 15, the mean premorbid IQ estimate for schizophrenia samples was 94.7 (0.35 standard deviations below the mean and at the lower end of the average range). In the figure, level 1 includes all studies that used recent diagnostic criteria (RDC, ICD-10, or DSM-III or later) while level 2 includes all studies that used recent diagnostic criteria (RDC, ICD-10, or DSM-III or later) and "short" IQ estimates; Level 3 includes all studies that used older diagnostic systems (pre-DSM-III or RDC) and "short" IQ estimates. (Woodberry KA et al. Premorbid IQ in Schizophrenia: A Meta-Analytic Review Am J Psychiatry 2008; 165:579-587). The term effect size corresponds to





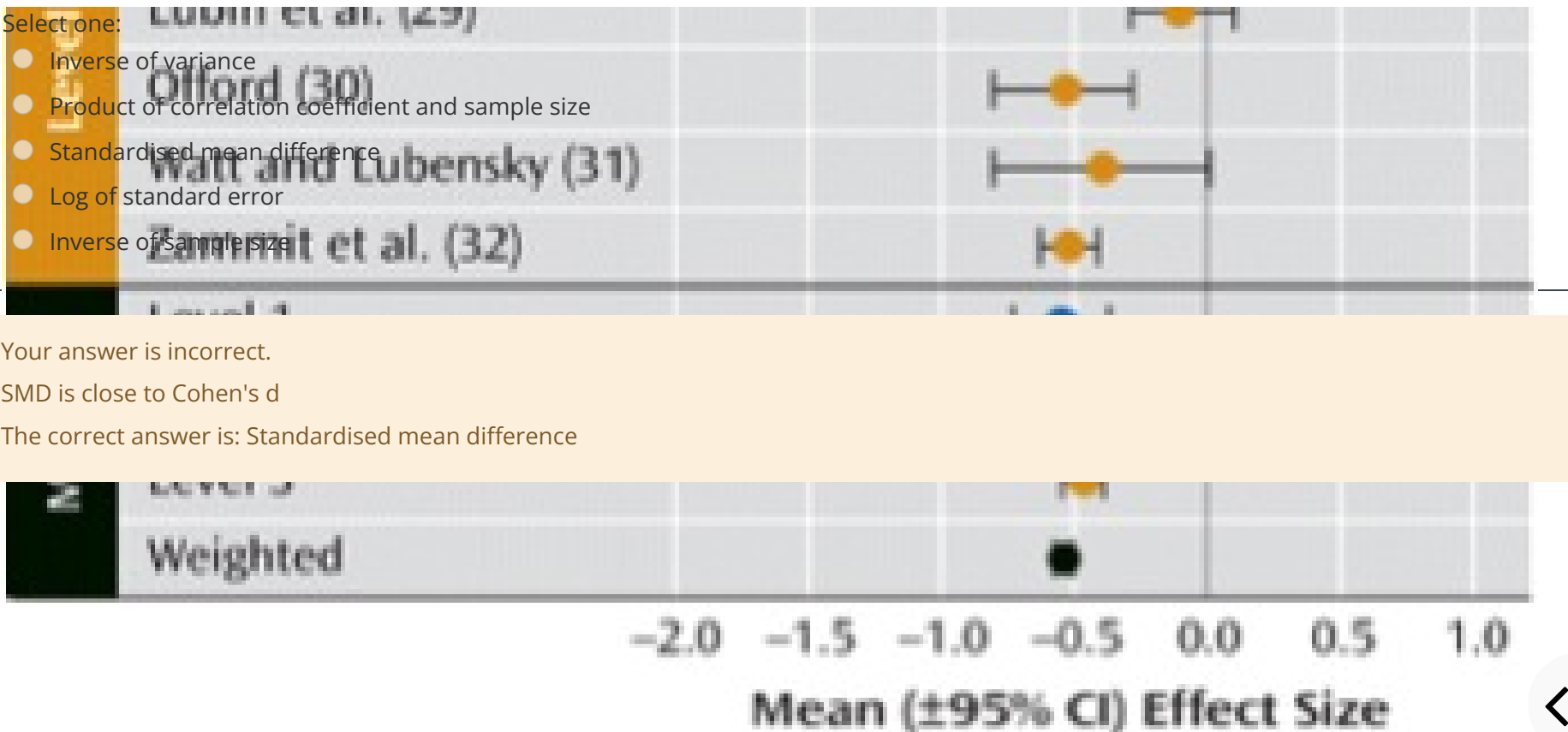
Select one:

- ☐ Inverse of variance
- ☐ Product of correlation coefficient and sample size
- ☐ Standardised mean difference
- ☐ Log of standard error
- ☐ Inverse of sample size

Your answer is incorrect.

SMD is close to Cohen's d

The correct answer is: Standardised mean difference



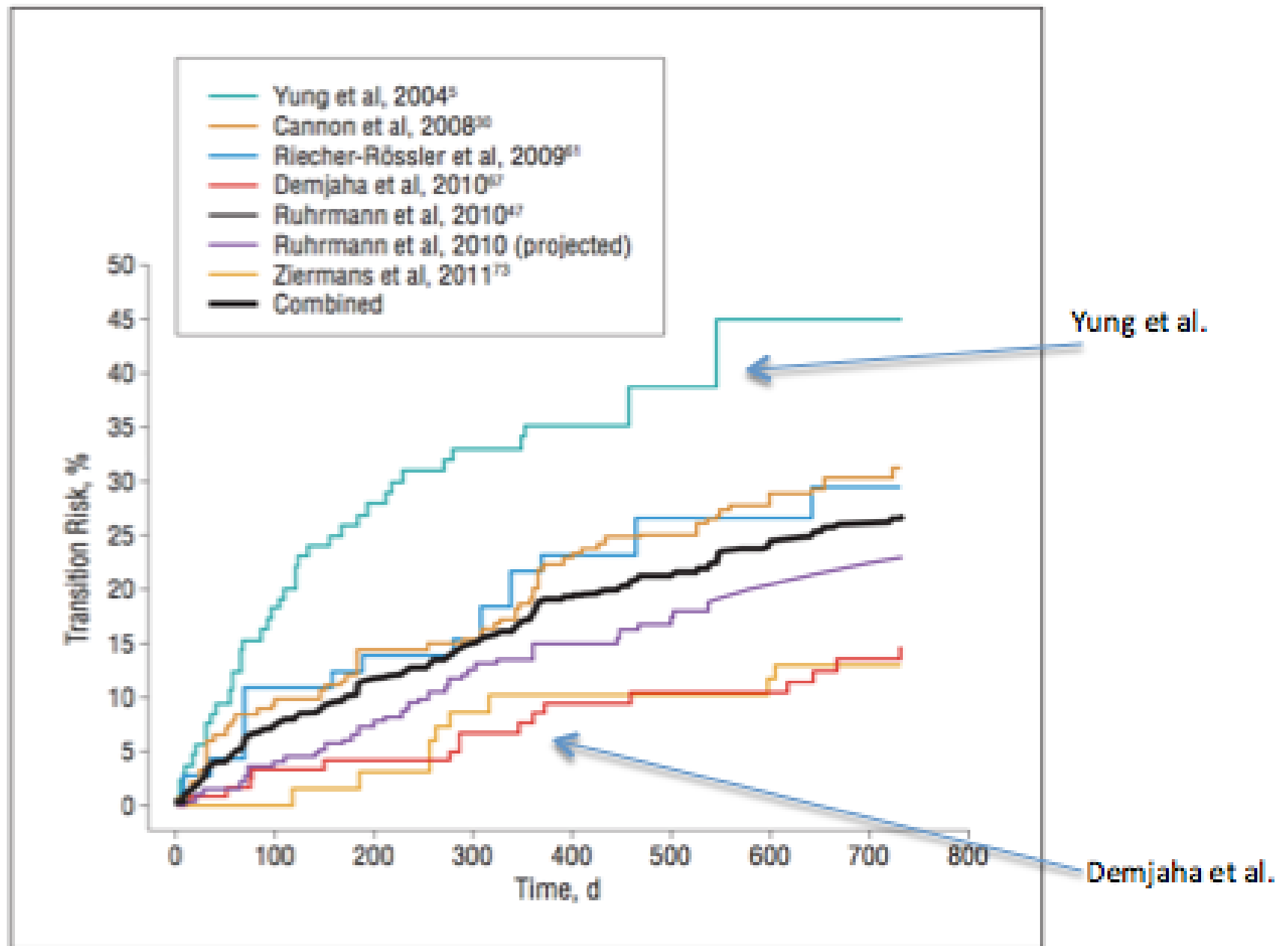
Question **26**

Not answered

Marked out of 1

To quantitatively examine the literature to date reporting the transition risk to psychosis in subjects at clinical high risk, electronic databases were searched until January 2011 and all studies reporting transition risks in patients at clinical high risk were retrieved. Twenty-seven studies met the inclusion criteria, comprising a total of 2502 patients. Psychosis transition was defined according to "standard" criteria (from the two major psychiatric diagnostic guidelines. The High-Risk state was defined using either the Ultra High-Risk criteria (MELBOURNE) or the Basic Symptoms criteria (BONN) using instruments such as the CAARMS, the Basel Screening Instrument for Psychosis and SIPS. Across the overall database, the pooled mean transition risk to a full psychotic episode from an HR state was 29.2% (95% CI, 27.3%-31.1%), with a mean follow-up of 31 months. This risk was independent of the psychometric instruments used and was estimated to be 18% after 6 months of follow-up, 22% after 1 year, 29% after 2 years, and 36% after 3 years. To provide a visual representation of the time course, a subset of studies (n = 6) with necessary data are summarized in the below Figure. (Fusar-Poli et al... Arch Gen Psychiatry. 2012;69(3):220-229). What type of graph is shown in the figure?





Select one:

- ☒ Kaplan-Meier curve (Fusar-Poli et al., Arch Gen Psychiatry, 2012;69(3):220-229)
- ☐ Scalogram curve
- ☐ Cox curve
- ☐ Meta-regression curve
- ☐ ROC curve

Your answer is incorrect.

This is a Kaplan-Meier curve. You don't see the usual stepping down pattern, as the authors have chosen to plot the risk of the outcome per se rather than the risk of surviving the outcome, in the y-axis.

The correct answer is: Kaplan Meier curve



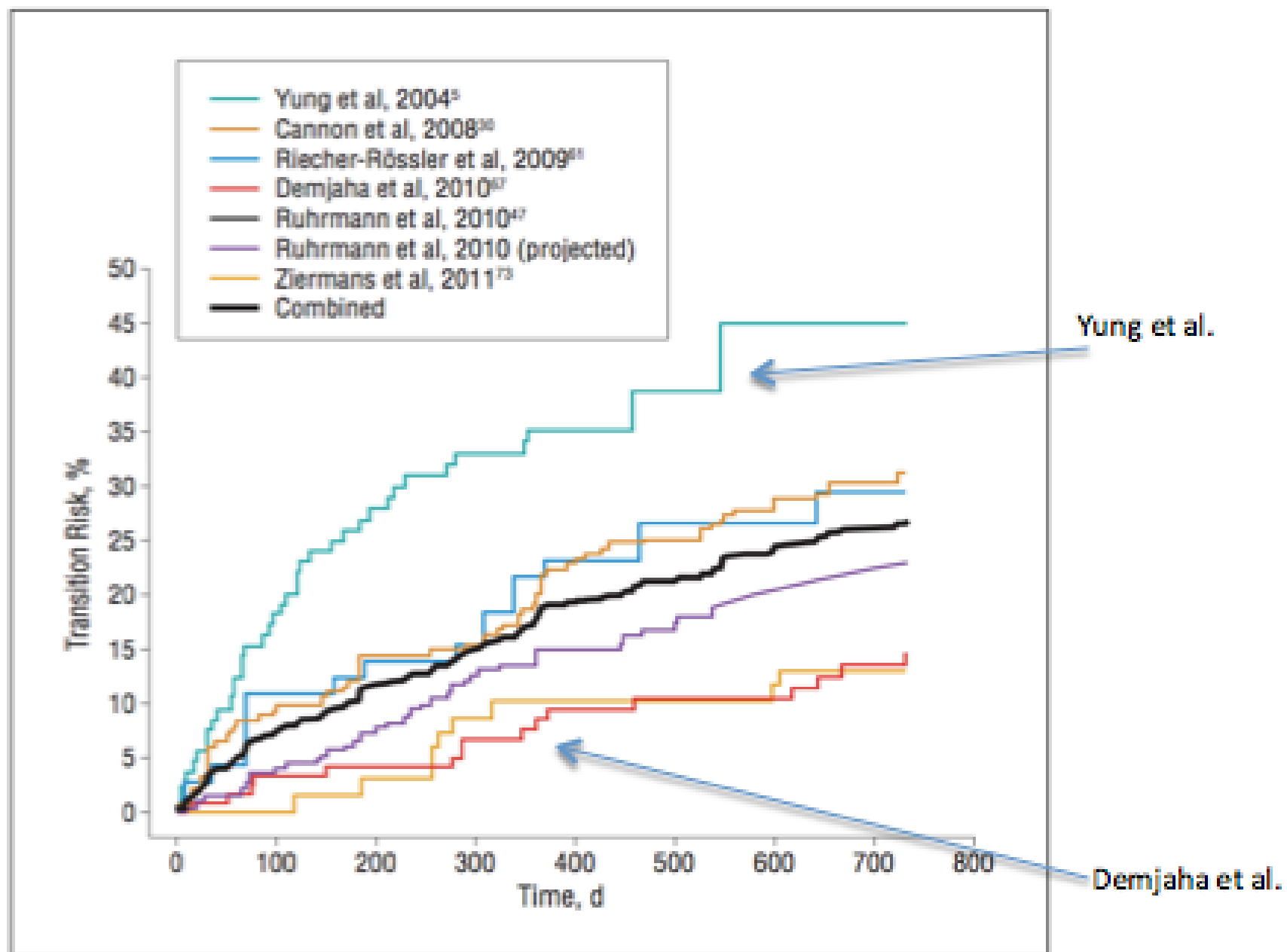
Question **27**

Not answered

Marked out of 1

To quantitatively examine the literature to date reporting the transition risk to psychosis in subjects at clinical high risk, electronic databases were searched until January 2011 and all studies reporting transition risks in patients at clinical high risk were retrieved. Twenty-seven studies met the inclusion criteria, comprising a total of 2502 patients. Psychosis transition was defined according to "standard" criteria (from the 2 major psychiatric diagnostic guidelines. The High-Risk state was defined using either the Ultra High-Risk criteria (MELBOURNE) or the Basic Symptoms criteria (BONN) using instruments such as the CAARMS, the Basel Screening Instrument for Psychosis and SIPS. Across the overall database, the pooled mean transition risk to a full psychotic episode from an HR state was 29.2% (95% CI, 27.3%-31.1%), with a mean follow-up of 31 months. This risk was independent of the psychometric instruments used and was estimated to be 18% after 6 months of follow-up, 22% after 1 year, 29% after 2 years, and 36% after 3 years. To provide a visual representation of the time course, a subset of studies (n = 6) with necessary data are summarized in the below Figure. (Fusar-Poli et al.. Arch Gen Psychiatry. 2012;69(3):220-229). The y-axis of this curve represents





Select one:

- ☐ The proportion of subjects with frank psychosis who were included in each study/sample
- ☐ The proportion of High Risk subjects who were successfully prevented from transition to frank psychosis
- ☐ The proportion of High Risk subjects who survived an episode of frank psychosis
- ☐ The proportion of High Risk subjects who developed a frank psychotic episode
- ☐ Follow-up time for individual studies

Your answer is incorrect.

Transition risk refers to the proportion of High-Risk Subjects who developed a frank psychotic episode. The authors have chosen to plot the risk of the outcome per se rather than the risk of surviving the outcome in the y-axis.

The correct answer is: The proportion of High Risk subjects who developed a frank psychotic episode



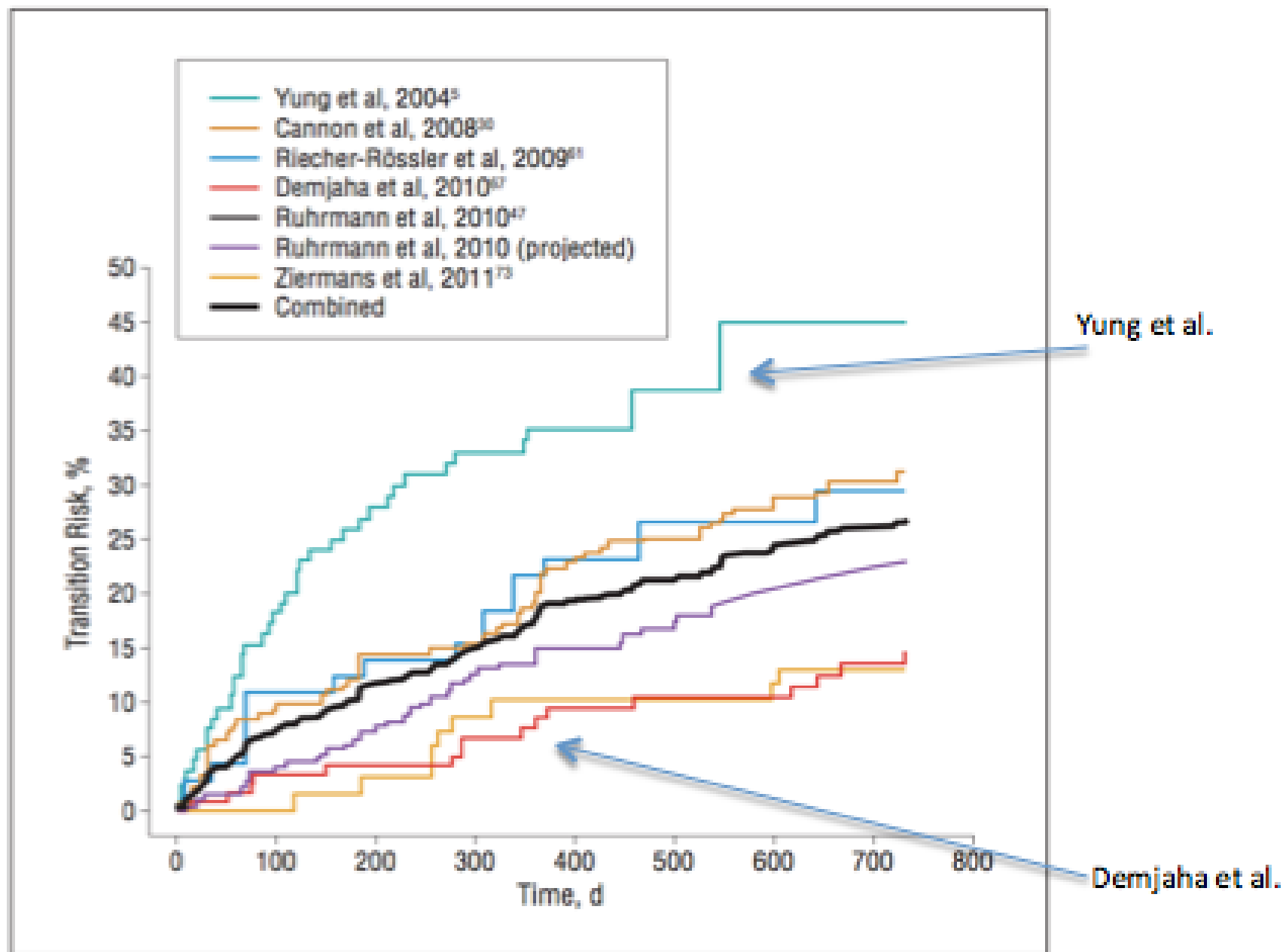
Question **28**

Not answered

Marked out of 1

To quantitatively examine the literature to date reporting the transition risk to psychosis in subjects at clinical high risk, electronic databases were searched until January 2011 and all studies reporting transition risks in patients at clinical high risk were retrieved. Twenty-seven studies met the inclusion criteria, comprising a total of 2502 patients. Psychosis transition was defined according to "standard" criteria (from the 2 major psychiatric diagnostic guidelines. The High-Risk state was defined using either the Ultra High-Risk criteria (MELBOURNE) or the Basic Symptoms criteria (BONN) using instruments such as the CAARMS, the Basel Screening Instrument for Psychosis and SIPS. Across the overall database, the pooled mean transition risk to a full psychotic episode from an HR state was 29.2% (95% CI, 27.3%-31.1%), with a mean follow-up of 31 months. This risk was independent of the psychometric instruments used and was estimated to be 18% after 6 months of follow-up, 22% after 1 year, 29% after 2 years, and 36% after 3 years. To provide a visual representation of the time course, a subset of studies (n = 6) with necessary data are summarized in the below Figure. (Fusar-Poli et al... Arch Gen Psychiatry. 2012;69(3):220-229). What type of study has been conducted?





Select one:

- ☒ A prospective cohort study (Fusar-Poli et al., Arch Gen Psychiatry, 2012;69(3):220-229)
- ☐ A cross-sectional study
- ☐ A retrospective cohort study
- ☐ A case-control study
- ☐ A meta-analytical study

Your answer is incorrect.

In general you see Kaplan-Meier curves in cohort studies. But, in this case, a meta-analysis has been conducted.

The correct answer is: A meta-analytical study



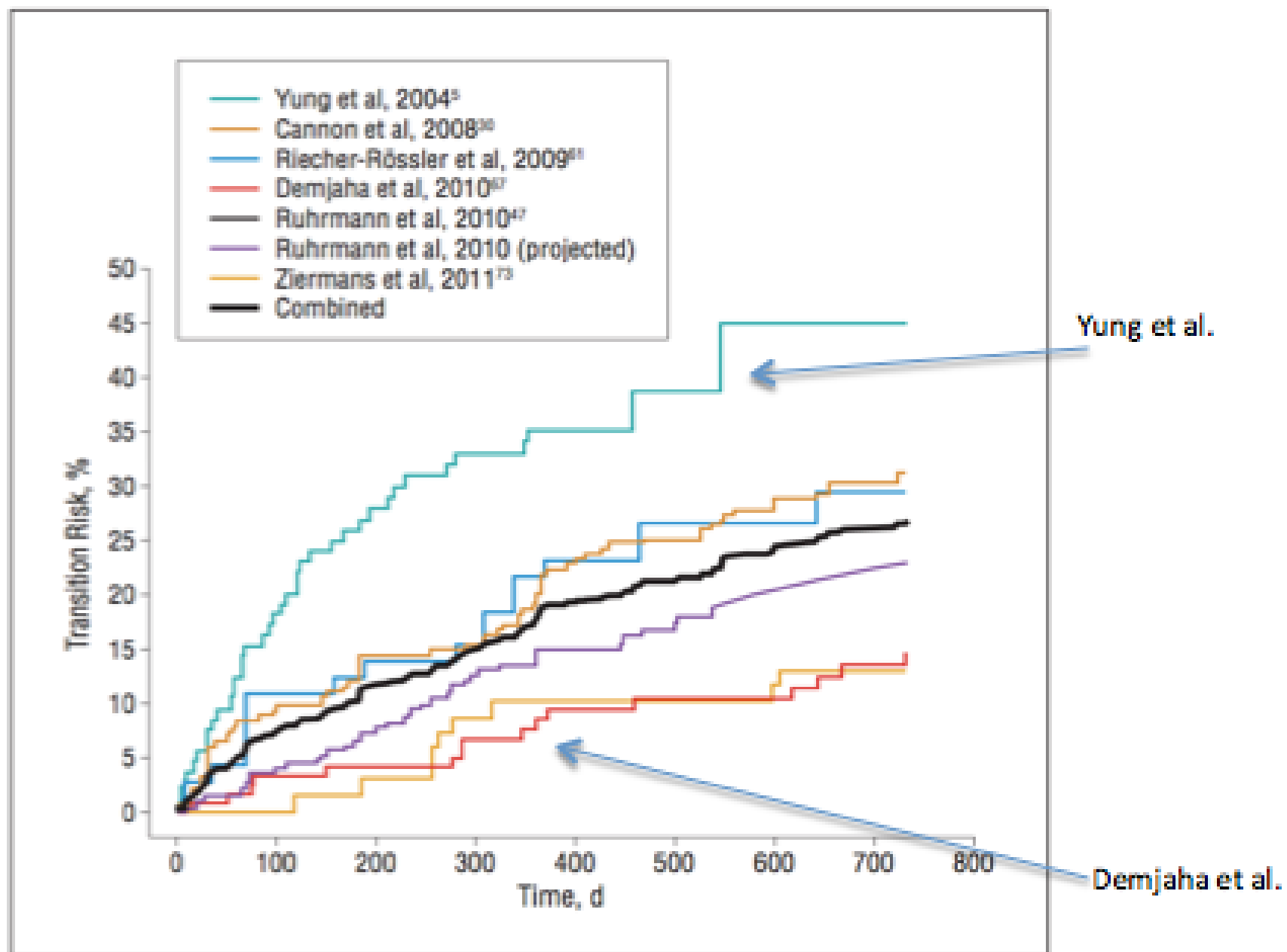
Question **29**

Not answered

Marked out of 1

To quantitatively examine the literature to date reporting the transition risk to psychosis in subjects at clinical high risk, electronic databases were searched until January 2011 and all studies reporting transition risks in patients at clinical high risk were retrieved. Twenty-seven studies met the inclusion criteria, comprising a total of 2502 patients. Psychosis transition was defined according to "standard" criteria (from the 2 major psychiatric diagnostic guidelines. The High-Risk state was defined using either the Ultra High-Risk criteria (MELBOURNE) or the Basic Symptoms criteria (BONN) using instruments such as the CAARMS, the Basel Screening Instrument for Psychosis and SIPS. Across the overall database, the pooled mean transition risk to a full psychotic episode from an HR state was 29.2% (95% CI, 27.3%-31.1%), with a mean follow-up of 31 months. This risk was independent of the psychometric instruments used and was estimated to be 18% after 6 months of follow-up, 22% after 1 year, 29% after 2 years, and 36% after 3 years. To provide a visual representation of the time course, a subset of studies (n = 6) with necessary data are summarized in the below Figure. (Fusar-Poli et al.. Arch Gen Psychiatry. 2012;69(3):220-229). The x-axis of this curve represents





Select one:

- ☐ Follow-up time after recruiting high-risk subjects
- ☐ Time to respond to preventive interventions in high-risk subjects
- ☐ Follow-up time after developing frank psychosis
- ☐ Time to develop high-risk features
- ☐ Time from point of transition to seeking treatment for psychosis

Your answer is incorrect.

The follow-up period employed in the individual studies is plotted along the x-axis. This time refers to the time for transition i.e. the time taken for frank psychosis to appear after being recruited into a study. In some cases, this may be equivalent to the time taken for frank psychosis to appear after developing high-risk features though this information is not available from the precis or the figure.

The correct answer is: Follow-up time after recruiting high-risk subjects



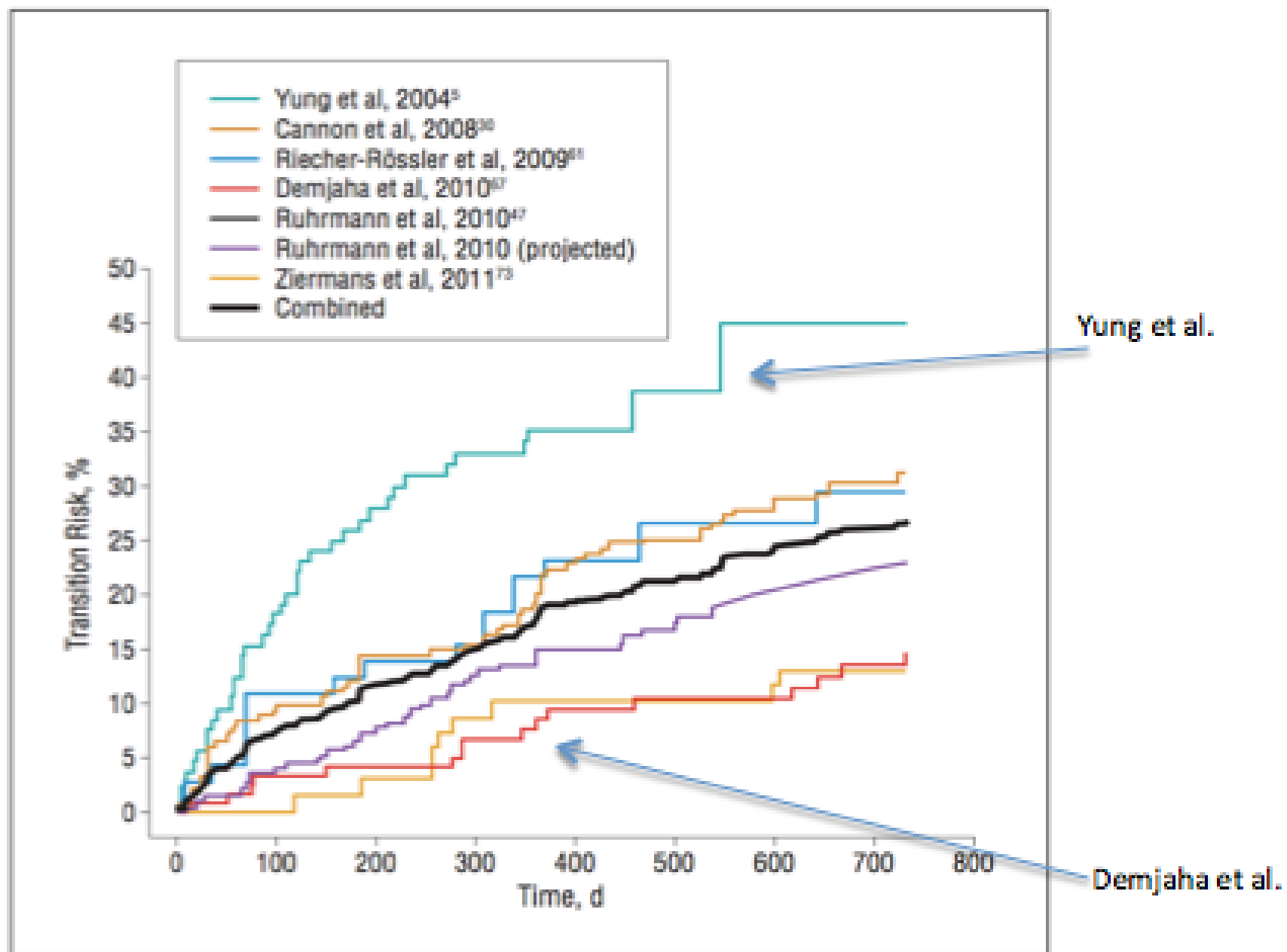
Question **30**

Not answered

Marked out of 1

To quantitatively examine the literature to date reporting the transition risk to psychosis in subjects at clinical high risk, electronic databases were searched until January 2011 and all studies reporting transition risks in patients at clinical high risk were retrieved. Twenty-seven studies met the inclusion criteria, comprising a total of 2502 patients. Psychosis transition was defined according to "standard" criteria (from the 2 major psychiatric diagnostic guidelines. The High-Risk state was defined using either the Ultra High-Risk criteria (MELBOURNE) or the Basic Symptoms criteria (BONN) using instruments such as the CAARMS, the Basel Screening Instrument for Psychosis and SIPS. Across the overall database, the pooled mean transition risk to a full psychotic episode from an HR state was 29.2% (95% CI, 27.3%-31.1%), with a mean follow-up of 31 months. This risk was independent of the psychometric instruments used and was estimated to be 18% after 6 months of follow-up, 22% after 1 year, 29% after 2 years, and 36% after 3 years. To provide a visual representation of the time course, a subset of studies (n = 6) with necessary data are summarized in the attached Figure. (Fusar-Poli et al.. Arch Gen Psychiatry. 2012;69(3):220-229). The median survival time of combined results is





Select one:

- ☐ 400 days (Fusar-Poli et al., Arch Gen Psychiatry, 2012;69(3):220-229)
- ☐ 500 days
- ☐ 300-400 days
- ☐ <300days
- ☐ Cannot be determined

Your answer is incorrect.

The time point at which the value on y-axis i.e. total transition rate (or the probability of survival in 'step-down' curves) is 50% is called median survival time. None of the individual curves met that point. The combined curve also does not meet 50% transition even at the end of 750 days period of follow-up.

The correct answer is: Cannot be determined



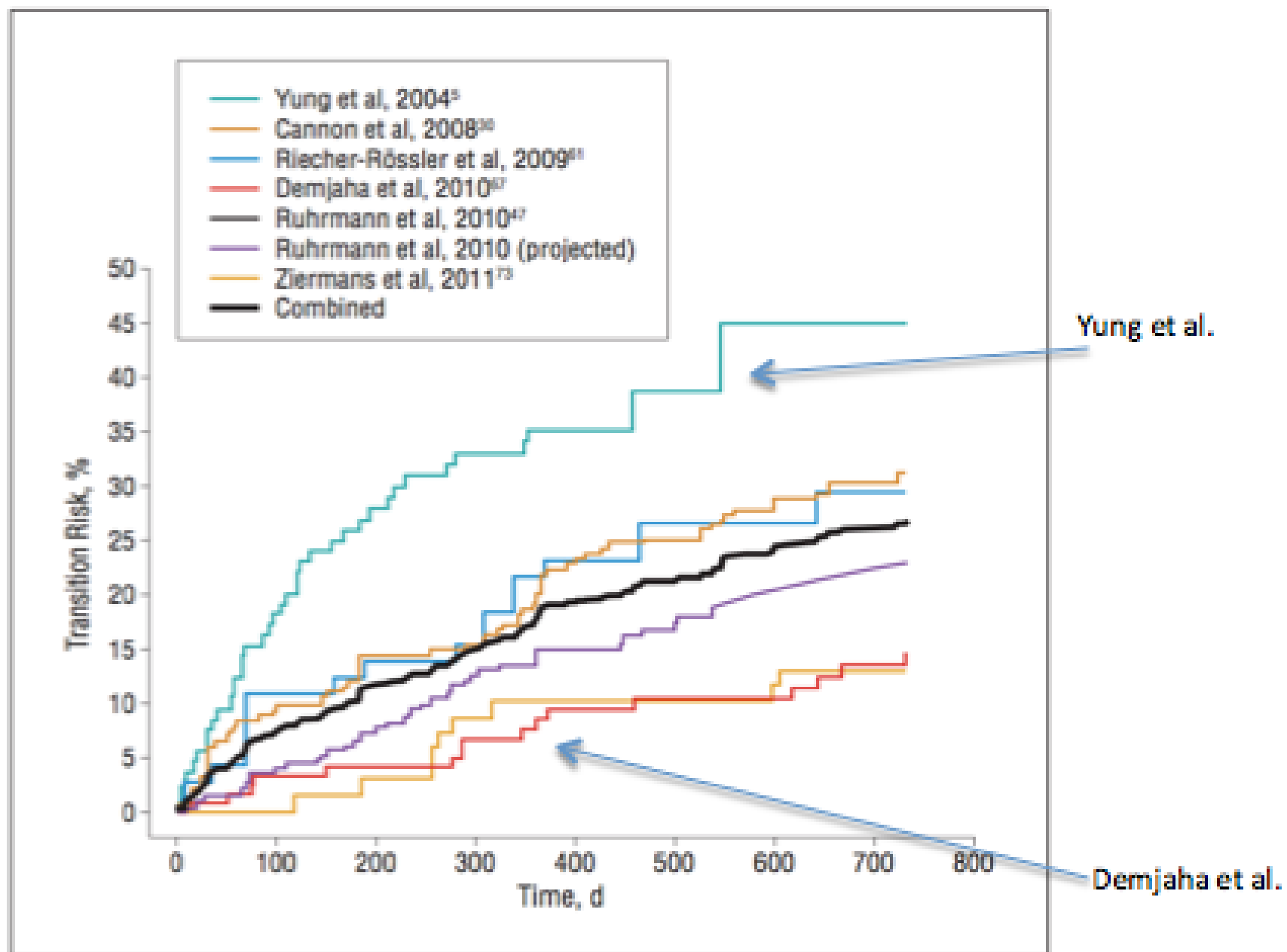
Question **31**

Not answered

Marked out of 1

To quantitatively examine the literature to date reporting the transition risk to psychosis in subjects at clinical high risk, electronic databases were searched until January 2011 and all studies reporting transition risks in patients at clinical high risk were retrieved. Twenty-seven studies met the inclusion criteria, comprising a total of 2502 patients. Psychosis transition was defined according to "standard" criteria (from the two major psychiatric diagnostic guidelines. The High-Risk state was defined using either the Ultra High-Risk criteria (MELBOURNE) or the Basic Symptoms criteria (BONN) using instruments such as the CAARMS, the Basel Screening Instrument for Psychosis and SIPS. Across the overall database, the pooled mean transition risk to a full psychotic episode from an HR state was 29.2% (95% CI, 27.3%-31.1%), with a mean follow-up of 31 months. This risk was independent of the psychometric instruments used and was estimated to be 18% after 6 months of follow-up, 22% after 1 year, 29% after 2 years, and 36% after 3 years. To provide a visual representation of the time course, a subset of studies (n = 6) with necessary data are summarized in the below Figure. (Fusar-Poli et al.. Arch Gen Psychiatry. 2012;69(3):220-229). With respect to the transition rates shown in the figure which of the following statements is correct?





Select one:

- ☐ After 500 days of follow-up, no frank psychosis occurred in the study reported by Demjaha et al.
- ☐ Nearly 75% of High-Risk subjects will develop frank psychosis by 700 days of follow-up
- ☐ At 400 days of follow-up the transition rate varies between <10% and 35%
- ☐ The lowest rate of transition was noted for the study reported by Yung et al.
- ☐ The range of transition risk decreases with increasing follow-up duration

Your answer is incorrect.

At 400 days of follow-up, the transition rate varies between <10% and 35%. Nearly 25% of High-Risk subjects will develop frank psychosis by 700 days of follow-up (as shown by the combined results). The range of transition risk increases with increasing follow-up duration. Yung et al. reported the highest rate of transition. Transition to frank psychosis started before 100 days of follow-up in most studies (except Demjaha et al.) and continued after that. Even after 500 days of follow-up, the risk of transition still continued with many new cases of frank psychosis occurring beyond this time in all studies (including that of Demjaha et al.).

The correct answer is: At 400 days of follow-up the transition rate varies between <10% and 35%



Question **32**

Not answered

Marked out of 1

Which one among the following is essentially an observational study of the evidence that allows large quantities of information to be speedily refined and reduced to a manageable size?

Select one:

- ☐ Systematic reviews
- ☐ Randomized clinical trials
- ☐ Clinical audit
- ☐ Economic analyses
- ☐ Qualitative studies

Your answer is incorrect.

A systematic review is essentially an observational study of the evidence that allows large quantities of information to be speedily refined and reduced to a manageable size. It is a scientific evaluation of several studies that have been conducted on a specific clinical question. By pooling together, the evidence drawn from systematic reviews can be very powerful and influential indeed. (Ref: Making sense of critical appraisal: Ajetunmobi- Pg 154)

The correct answer is: Systematic reviews

Question **33**

Not answered

Marked out of 1

Which one among the following is not a procedure followed when conducting a systematic review?

Select one:

- ☐ Critical appraisal of evidence
- ☐ Synthesis of evidence
- ☐ Specification of evidence sought
- ☐ Randomisation
- ☐ Identification of evidence

Your answer is incorrect.

The typical stages in a systematic overview include 1) specification of evidence sought 2) identification of evidence 3) critical appraisal of evidence 4) synthesis of evidence (Higgins 2008). Randomisation is not a step involved in this process.

The correct answer is: Randomisation



Question **34**

Not answered

Marked out of 1

Which of the following statements is inaccurate about risk ratios in meta-analytical studies?

Select one:

- ☐ Risk ratio is calculated by dividing the risk in the treated group by the risk in the control group
- ☐ Risk ratio of less than 1 indicates no difference between the groups
- ☐ Risk ratios are frequently given with their confidence intervals
- ☐ Risk ratios are used to investigate the effects of a treatment
- ☐ If the confidence interval does not contain 1, it is statistically significant

Your answer is incorrect.

Risk ratios are used to investigate the effects of a treatment. The risk ratio is calculated by dividing the risk in the treated group by the risk in the control group. A risk ratio of 1 indicates no difference between the groups. Risk ratios are frequently given with their confidence intervals. If the confidence interval does not contain 1 (no difference in risk), it is statistically significant.

The correct answer is: Risk ratio of less than 1 indicates no difference between the groups



Question **35**

Not answered

Marked out of 1

Which of the following statements is inaccurate regarding search strategy for systematic reviews?

Select one:

- ☐ Searching one electronic database is insufficient
- ☐ It is important to focus on peer-reviewed published studies
- ☐ It is important to search for studies in all languages, not just English
- ☐ The search strategy used needs to be exhaustive
- ☐ The search strategy must be comprehensive

Your answer is incorrect.

It is important to search for both published and unpublished studies, in all languages, not just English. The search strategy used needs to be comprehensive and exhaustive using different electronic databases. All these measures could help to avoid publication bias.

The correct answer is: It is important to focus on peer-reviewed published studies



Question **36**

Not answered

Marked out of 1

Which one among the following is not essential for conducting a meta-analysis?

Select one:

- ☐ Exhaustive literature search
- ☐ Recording of data from the individual studies
- ☐ Establishing criteria for including and excluding studies
- ☐ Researcher being an expert in the field of inquiry
- ☐ Statistical analysis of the data.

Your answer is incorrect.

The four basic steps for conducting a meta-analysis include (1) Literature search, (2) Establishing criteria for including and excluding studies (3) Recording of data from the individual studies, and (4) Statistical analysis of the data.

The correct answer is: Researcher being an expert in the field of inquiry



Question **37**

Not answered

Marked out of 1

Which plot can be used to detect publication bias?

Select one:

- ☐ Kaplan-Meier plot
- ☐ Scatter plot
- ☐ Dot plot
- ☐ Funnel plot
- ☐ ROC plot

Your answer is incorrect.

Publication bias can be detected by constructing a funnel plot of all identifiable studies that have been performed on a clinical subject. A funnel plot involves plotting the study size against the odds ratio. Any biases in reporting will skew the normally symmetrical shape of the plot.

The correct answer is: Funnel plot

Question 38

Not answered

Marked out of 1

While conducting a systematic review, publication bias could be determined using all of the following methods except

Select one:

- ☐ Confidence interval of the pooled effect size
- ☐ Failsafe N
- ☐ Soliciting and comparing published vs. unpublished data
- ☐ Galbraith plot
- ☐ Funnel plot

Your answer is incorrect.

Publication bias can be detected using all of the various listed methods except using the confidence interval of the pooled estimate. The confidence interval of the pooled estimate is a function of the total sample size and the number of trials included in a meta-analysis.

The correct answer is: Confidence interval of the pooled effect size



© 2023 SPMM Course Limited, UK.

[Refund Policy](#) | [Terms & Conditions](#)

